Exercises on compressed sensing

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1 Complexity

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Exercise 1.1 (Volumetric argument)
Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. Denote by B its unit ball. Let $\varepsilon > 0$ and $\Lambda \subset B$ be such that for all $x, y \in \Lambda$, if $x \neq y$ then

$$\|x - y\| \geq \varepsilon. \tag{1}$$

Then the cardinality of Λ necessarily satisfies

$$|\Lambda| \leq \Big(1 + \frac{2}{\varepsilon}\Big)^n.$$

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Solution of the exercise 1.1 One has

$$\bigcup_{x\in\Lambda} \big(x + (\varepsilon/2)\mathring{B}\big) \subset B + (\varepsilon/2)B = (1 + \varepsilon/2)B. \tag{2}$$

Moreover, the sets $(x + (\varepsilon/2)\mathring{B})_{x\in\Lambda}$ are disjoint. Taking volumes in (2) therefore yields

$$\sum_{x\in\Lambda} \mathrm{Vol}\big(x + (\varepsilon/2)\mathring{B}\big) \leq \mathrm{Vol}\big((1 + \varepsilon/2)B\big).$$

Volume is invariant under translation, so for every $x \in \Lambda$, $\mathrm{Vol}\big(x + (\varepsilon/2)\mathring{B}\big) = \mathrm{Vol}((\varepsilon/2)\mathring{B})$. By scaling, $\mathrm{Vol}((\varepsilon/2)\mathring{B}) = (\varepsilon/2)^n \mathrm{Vol}(\mathring{B})$ and $\mathrm{Vol}\big((1+\varepsilon/2)B\big) = (1+\varepsilon/2)^n \mathrm{Vol}(B)$. The boundary has measure zero, so $\mathrm{Vol}(B) = \mathrm{Vol}(\mathring{B})$. Hence

$$|\Lambda|(\varepsilon/2)^n \mathrm{Vol}(B) \leq (1+\varepsilon/2)^n \mathrm{Vol}(B).$$

Thus $|\Lambda| \leq (1+2/\varepsilon)^n$.

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Exercise 1.2 (Varshamov–Gilbert)
Let $s \leq N/2$. There exists a family $\mathcal{S}$ of subsets of $\{1,\dots,N\}$ such that

1. **for every $S \in \mathcal{S}$, $|S| = s$,**
2. **for all $S_1, S_2 \in \mathcal{S}$, $S_1 \neq S_2 \Rightarrow |S_1 \cap S_2| \leq \lfloor s/2 \rfloor$,**
3. **$\log|\mathcal{S}| \geq \lfloor \frac{s}{2} \rfloor \log\Big(\lfloor \frac{N}{8es} \rfloor\Big)$.**

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Solution of the exercise 1.2 Enumerate successively the subsets of cardinality s of $\{1,\dots,N\}$ and discard those that are not wanted. Without loss of generality, assume that $s/2$ is an integer.

Let S_1 be a subset of $\{1,...,N\}$ of cardinality s. Discard every subset J of $\{1,...,N\}$ of cardinality s such that $|S_1 \cap J| > s/2$. There are

$$N_s := \sum_{k=s/2+1}^{s} \binom{s}{k}\binom{N-s}{s-k}$$

of them, and since $s \leq N/2$,

$$\sum_{k=s/2+1}^{s} \binom{s}{k}\binom{N-s}{s-k} \leq 2^s \max_{s/2+1\leq k\leq s} \binom{N-s}{s-k} \leq 2^s \binom{N}{s/2}.$$

Indeed,

$$\sum_{k=0}^{s} \binom{s}{k} = 2^s$$

and for every $s/2+1 \leq k \leq s$, splitting the product gives

$$\begin{aligned}
\binom{N}{s/2} &= \frac{N}{s/2}\cdot\frac{N-1}{s/2-1}\cdots\frac{N-(s/2-1)}{1} \\
&= \underbrace{\frac{N}{s/2}}_{\geq 1}\cdot\underbrace{\frac{N-1}{s/2-1}}_{\geq 1}\cdots\underbrace{\frac{N-(s/2-s+k-1)}{s-k+1}}_{\geq 1}\times\underbrace{\frac{N-(s/2-s+k)}{s-k}}_{\geq \frac{N-s}{s-k}}\cdots\underbrace{\frac{N-(s/2-1)}{1}}_{\geq \frac{N-s-(s-k-1)}{1}} \\
&\geq \frac{N-s}{s-k}\cdots\frac{N-s-(s-k-1)}{1} = \binom{N-s}{s-k}
\end{aligned}$$

where $N \geq s/2$ is used to bound the first factors from below by 1, and $k \leq s \leq 3s/2$ is used for the remaining factors. One also uses $k \geq s/2$ to see that $\binom{N}{s/2}$ has more factors ($s/2$ of them) than $\binom{N-s}{s-k}$ (which has $s-k \leq s/2$ factors).

Choose S_2 of cardinality s among the remaining sets, of which there are

$$\binom{N}{s} - N_s.$$

As before, discard every remaining subset of size s that shares more than $s/2$ elements with S_2. At most N_s sets are removed (not exactly N_s, since some sets that intersect S_2 in more than $s/2$ elements may already have been removed because they intersected S_1 in more than $s/2$ elements).

At the next step, choose $S_3 \subset \{1, \dots, N\}$ of cardinality s among at least $\binom{N}{s} - 2N_s$ sets. Iterate p times until

$$\binom{N}{s} - pN_s > 0 \text{ and } \binom{N}{s} - (p+1)N_s \leq 0,$$

that is

$$p = \left\lfloor \binom{N}{s} \Big/ N_s \right\rfloor \geq \left\lfloor \binom{N}{s} \Big/ \left(2^s \binom{N}{s/2}\right) \right\rfloor.$$

This produces a family $\mathcal{S} = \{S_1, S_2, \dots, S_p\}$ of subsets of cardinality s that are mutually s-separated for the Hamming distance ($d(S_1, S_2) = |S_1 \Delta S_2| = |S_1 \cup S_2| - |S_1 \cap S_2|$) and such that

$$|\mathcal{S}| \geq \left\lfloor \binom{N}{s} \Big/ \left(2^s \binom{N}{s/2}\right) \right\rfloor.$$

Since $s \leq N/2$, one has $\left(\frac{N}{s}\right)^s \leq \binom{N}{s} \leq \left(\frac{eN}{s}\right)^s$ (see the exercise *Counting and concentration* in Chapter 2), and therefore

$$|\mathcal{S}| \geq \left\lfloor \frac{(N/s)^s}{2^s (eN/(s/2))^{(s/2)}} \right\rfloor \geq \left\lfloor \left(\frac{N}{8es}\right)^{s/2} \right\rfloor \geq \left\lfloor \frac{N}{8es} \right\rfloor^{\lfloor s/2 \rfloor},$$

which concludes the proof.

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Exercise 1.3 (Computing the Dudley integral of B_1^N with respect to ℓ_2^N)

Consider the unit balls $B_1^N = \{x \in \mathbb{R}^N : \sum_{i=1}^N |x_i| \leq 1\}$ and $B_2^N = \{x \in \mathbb{R}^N : \sum_{i=1}^N x_i^2 \leq 1\}$. For every $\varepsilon > 0$, write $N(B_1^N, \varepsilon B_2^N)$ for the smallest number of translates of the ball εB_2^N needed to cover B_1^N. A measure of the complexity of B_1^N for the metric ℓ_2^N is the Dudley integral

$$I(B_1^N, \ell_2^N) := \int_0^\infty \sqrt{\log N(B_1^N, \varepsilon B_2^N)} d\varepsilon.$$

We will show that there exists an absolute constant $c_0 > 0$ such that $I(B_1^N, \ell_2^N) \leq c_0 (\log N)^{3/2}$, by a probabilistic argument.

First compute the entropy of B_1^N with respect to ℓ_2^N. Let $\varepsilon > 0$. The goal is to build an ε-net of B_1^N for ℓ_2^N: a smallest possible set of points $x_1, \dots, x_p \in B_1^N$ such that for every $x \in B_1^N$ there exists $i_0 \in \{1, \dots, p\}$ with $\|x - x_{i_0}\|_2 \leq \varepsilon$. Let $x \in B_1^N$, and write $x = \sum_{i=1}^N \lambda_i e_i$

where $(e_1, \dots, e_N)$ is the canonical basis of $\mathbb{R}^N$ and $\sum_{i=1}^N |\lambda_i| \leq 1$. Consider the random variable X with values in $\{\pm e_1, \dots, \pm e_N, 0\}$ defined by

$$\mathbb{P}[X = \mathrm{Sign}(\lambda_i) e_i] = |\lambda_i|, \, \forall i = 1, \dots, N \text{ and } \mathbb{P}[X = 0] = 1 - \|x\|_1 .$$

1. **Compute the expectation of X.**

2. **Let n be an integer to be chosen later, and let $X_1, \dots, X_n$ be i.i.d. copies of X. Show that $\mathbb{E} \left\| n^{-1} \sum_{i=1}^n X_i - \mathbb{E}X \right\|_2^2 \leq n^{-1}$.**

3. **Deduce that if $n = \lceil \varepsilon^{-2} \rceil$, then**

$$\Lambda_\varepsilon := \Big\{ \frac{1}{n} \sum_{i=1}^n z_i : z_1, \dots, z_n \in \{\pm e_1, \dots, \pm e_N, 0\} \Big\}$$

   is an ε-net of B_1^N with respect to ℓ_2^N. Deduce a bound on $\log N(B_1^N, \varepsilon B_2^N)$.

4. **Show that for every $\eta \geq \varepsilon$,**

$$\log N(B_1^N, \varepsilon B_2^N) \leq \log N(B_1^N, \eta B_2^N) + N \log \Big(\frac{3\eta}{\varepsilon} \Big).$$

5. **Deduce that for every $0 < \varepsilon \leq N^{-1/2}$,**

$$\log N(B_1^N, \varepsilon B_2^N) \leq c_0 N \log \big(c_1 / (N\varepsilon^2) \big).$$

6. **Finally, prove that**

$$\log N(B_1^N, \varepsilon B_2^N) \leq c_0 \begin{cases} 0 & \text{when } \varepsilon \geq 1, \\ \frac{\log(eN)}{\varepsilon^2} & \text{when } N^{-1/2} \leq \varepsilon \leq 1, \\ N \log \left(\frac{e}{N\varepsilon^2} \right) & \text{when } 0 < \varepsilon \leq N^{-1/2}. \end{cases}$$

   Conclude that there exists $c_0 > 0$ such that $I(B_1^N, \ell_2^N) \leq c_0 (\log N)^{3/2}$.

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Solution of the exercise 1.3

1.
$$\mathbb{E}X = \sum_{y \in \{\pm e_1, \dots, \pm e_N, 0\}} y \mathbb{P}[X = y] = \sum_{i=1}^N \mathrm{Sign}(\lambda_i) e_i |\lambda_i| = x.$$

2. One has
$$\mathbb{E} \left\| \frac{1}{n} \sum_{i=1}^n X_i - \mathbb{E}X \right\|_2^2 = \frac{\mathbb{E} \|X - \mathbb{E}X\|_2^2}{n} = \frac{\mathbb{E} \|X\|_2^2 - \|\mathbb{E}X\|_2^2}{n} = \frac{\|x\|_1 - \|x\|_2^2}{n} \leq \frac{1}{n}.$$

3. We have seen that $\mathbb{E}X = x$ and $\mathbb{E}\left\|n^{-1}\sum_{i=1}^n X_i - \mathbb{E}X\right\|_2^2 \le \varepsilon^2$, so there exists $\omega \in \Omega$ such that

$$\left\| n^{-1}\sum_{i=1}^n X_i(\omega) - x \right\|_2 \le \varepsilon.$$

And $n^{-1}\sum_{i=1}^n X_i(\omega) \in \Lambda_\varepsilon$, so there exists $y \in \Lambda_\varepsilon$ with $\|y - x\|_2 \le \varepsilon$. Since $x \in B_1^N$ was arbitrary, Λ_ε is an ε-net of B_1^N for ℓ_2^N.

Thus $N(B_1^N, \varepsilon B_2^N)$ is at most the cardinality of Λ_ε. Since $|\Lambda_\varepsilon| \le (2N+1)^n$, one has $\log N(B_1^N, \varepsilon B_2^N) \le n\log(2N+1)$. For $\varepsilon \le 1$, $n = \lceil \varepsilon^{-2} \rceil \le 2\varepsilon^{-2}$, so $\log N(B_1^N, \varepsilon B_2^N) \le 2\varepsilon^{-2}\log(2N+1)$.

Note: In fact one can show that $|\Lambda_\varepsilon| \le (c_0 N/n + 1)^n$, and therefore $\log N(B_1^N, \varepsilon B_2^N) \lesssim \varepsilon^{-2}\log(c_0 N\varepsilon^2)$ when $N^{-1/2} \le \varepsilon \le 1$.

4. To cover B_1^N by balls εB_2^N, first cover B_1^N by balls ηB_2^N, then cover each ball ηB_2^N by balls εB_2^N. Then

$$N(B_1^N, \varepsilon B_2^N) \le N(B_1^N, \eta B_2^N) N(\eta B_2^N, \varepsilon B_2^N).$$

A volumetric argument also gives

$$N(\eta B_2^N, \varepsilon B_2^N) = N\Big(B_2^N, \frac{\varepsilon}{\eta} B_2^N\Big) \le \Big(1 + \frac{2\eta}{\varepsilon}\Big)^N.$$

5. By the note in 3), for $\eta = N^{-1/2}$ one has $n = \lceil \eta^{-2} \rceil = N$ and $|\Lambda_\eta| \le (c_0+1)^N$, so $\log N(B_1^N, \eta B_2^N) \le N\log(c_0+1)$. Question 4) then gives, for every $0 < \varepsilon \le N^{-1/2}$,

$$\log N(B_1^N, \varepsilon B_2^N) \le N\log(c_0+1) + N\log\Big(\frac{3}{\varepsilon\sqrt{N}}\Big) = \frac{N}{2}\log\Big(\frac{9(c_0+1)^2}{N\varepsilon^2}\Big) \le c_0' N \log\Big(\frac{c_1}{N\varepsilon^2}\Big),$$

up to increasing the absolute constants c_0' and c_1.

6. One has

$$I(B_1^N, \ell_2^N) \le \int_0^{N^{-1/2}} \sqrt{N\log\Big(\frac{e}{N\varepsilon^2}\Big)}\, d\varepsilon + \int_{N^{-1/2}}^1 \sqrt{\frac{\log(eN)}{\varepsilon^2}}\, d\varepsilon.$$

For the first integral, the change of variables $u = N\varepsilon^2$ gives $d\varepsilon = du/(2\sqrt{Nu})$ and

$$\int_0^{N^{-1/2}} \sqrt{N\log\Big(\frac{e}{N\varepsilon^2}\Big)}\, d\varepsilon = \frac{1}{2}\int_0^1 \sqrt{\frac{\log(e/u)}{u}}\, du,$$

which is a finite absolute constant. The second integral equals

$$\sqrt{\log(eN)}\int_{N^{-1/2}}^1 \frac{d\varepsilon}{\varepsilon} = \frac{1}{2}\sqrt{\log(eN)}\,\log N \le \frac{1}{2}\big(\log(eN)\big)^{3/2}.$$

Thus $I(B_1^N, \ell_2^N) \le c_0(\log(eN))^{3/2}$.

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Exercise 1.4 (Lower bound on the Dudley integral of B_1^N with respect to ℓ_2^N)
Using the Varshamov–Gilbert bound, show that

$$I(B_1^N, \ell_2^N) = \int_0^\infty \sqrt{\log N(B_1^N, \varepsilon B_2^N)} d\varepsilon \geq c_1 (\log N)^{3/2}$$

for some absolute constant $c_1 > 0$.

One may first show that if $N(B_1^N, \varepsilon B_2^N)$ is the smallest number of translates of εB_2^N needed to cover B_1^N, and $M(B_1^N, \varepsilon B_2^N)$ is the largest number of ε-separated points of B_1^N with respect to ℓ_2^N (that is, $\|x-y\|_2 \geq \varepsilon$), then

$$M(B_1^N, 2\varepsilon B_2^N) \leq N(B_1^N, \varepsilon B_2^N) \leq M(B_1^N, \varepsilon B_2^N).$$

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Solution of the exercise 1.4 Let Λ be a maximal ε-separated subset of B_1^N for ℓ_2^N, of cardinality $M(B_1^N, \varepsilon B_2^N)$. If $x \in B_1^N$, either $x \in \Lambda$, or $\Lambda \cup \{x\}$ is no longer ε-separated. In both cases there exists $y \in \Lambda$ with $\|x-y\|_2 \leq \varepsilon$. Thus Λ is an ε-net and

$$N(B_1^N, \varepsilon B_2^N) \leq M(B_1^N, \varepsilon B_2^N).$$

Conversely, two points at distance at least 2ε cannot lie in the same ball of radius ε (if the distance is exactly 2ε they lie on the boundary of a closed ball; replacing 2ε by 3ε only changes the absolute constant below). Hence

$$N(B_1^N, \varepsilon B_2^N) \geq M(B_1^N, 2\varepsilon B_2^N).$$

Let $1 \leq s \leq N/2$. By the Varshamov–Gilbert bound, there is a collection $\mathcal{S}$ of subsets of $\{1, \dots, N\}$ such that:

1. for every $S \in \mathcal{S}$, $|S| = s$;
2. for all $S_1 \neq S_2 \in \mathcal{S}$, $|S_1 \cap S_2| \leq s/2$;
3. $\log |\mathcal{S}| \geq \lfloor s/2 \rfloor \log\big(\lfloor N/(8es) \rfloor\big)$.

Set

$$\Lambda_s = \big\{x(S) : S \in \mathcal{S}\big\}, \qquad x(S) = \frac{1}{s} \sum_{j \in S} e_j,$$

where $(e_1, \dots, e_N)$ is the canonical basis of $\mathbb{R}^N$. For $S_1 \neq S_2$,

$$|S_1 \Delta S_2| = |S_1| + |S_2| - 2|S_1 \cap S_2| \geq s,$$

so

$$\|x(S_1) - x(S_2)\|_2 = \sqrt{|S_1 \Delta S_2|}/s \geq 1/\sqrt{s}.$$

Moreover $\|x(S)\|_1 = 1$, so $\Lambda_s \subset B_1^N$ is $(1/\sqrt{s})$-separated for ℓ_2^N and

$$\log M\big(B_1^N, s^{-1/2} B_2^N\big) \geq \log |\Lambda_s| \geq \frac{s}{2} \log\Big(\frac{N}{8es}\Big),$$

as soon as $N \geq 8es$ (otherwise the lower bound is not positive). For $\varepsilon = 1/(2\sqrt{s})$ one has $2\varepsilon = s^{-1/2}$ and $s = 1/(4\varepsilon^2)$, hence

$$\log N(B_1^N, \varepsilon B_2^N) \geq \log M(B_1^N, 2\varepsilon B_2^N) \geq \frac{1}{8\varepsilon^2} \log\Big(\frac{N\varepsilon^2}{2e}\Big).$$

The Dudley integral therefore satisfies, for an absolute constant $c > 0$,

$$I(B_1^N, \ell_2^N) \geq \int_{c/\sqrt{N}}^{1/2} \frac{\sqrt{\log(c_1 N\varepsilon^2)}}{\varepsilon}\, d\varepsilon.$$

The change of variables $u = \log(c_1 N\varepsilon^2)$ gives $du = 2\, d\varepsilon/\varepsilon$, and the integral is of order $(\log N)^{3/2}$.

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Exercise 1.5 (Complexity of the combinatorial cube for the Hamming distance)

Let N be an integer and let $\mathcal{C}_N = \{0,1\}^N$ be the combinatorial cube, equipped with the Hamming distance: for all $x, y \in \mathcal{C}_N$,

$$\rho(x,y) = \sum_{i=1}^{N} I(x_i \neq y_i)$$

where $I(x_i \neq y_i) = 1$ when $x_i \neq y_i$ and $I(x_i \neq y_i) = 0$ when $x_i = y_i$. Fix an integer $1 \leq k < N/2$. We are interested in subsets of $\mathcal{C}_N$ that are k-separated for the Hamming distance.

1. **Let X be uniformly distributed on $\mathcal{C}_N$. Let $x \in \mathcal{C}_N$. Show that**

$$2^N \mathbb{P}\big[\rho(X,x) \leq k\big] = |\{y \in \mathcal{C}_N : \rho(y,x) \leq k\}|$$

   where $|E|$ denotes the cardinality of a set E.

2. **Given $x \in \mathcal{C}_N$, determine the law of $\rho(X,x)$.**

3. **Given $x \in \mathcal{C}_N$, use 1), 2) and Chernoff's bound to upper bound the number of points at Hamming distance at most k from x.**

4. **Construct a set of k-separated points of cardinality at least**

$$\left\lfloor \frac{2^N - 1}{\big(eN/k\big)^k} \right\rfloor.$$

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Solution of the exercise 1.5

1. Since X is uniform on $\mathcal{C}_N$, $\mathbb{P}[X = y] = 2^{-N}$ for every $y \in \mathcal{C}_N$, and

$$\mathbb{P}\big[\rho(X,x) \leq k\big] = \sum_{y \in \mathcal{C}_N} I(\rho(y,x) \leq k)\mathbb{P}[X = y] = \frac{1}{2^N}|\{y \in \mathcal{C}_N : \rho(y,x) \leq k\}|.$$

2. Let $x \in \mathcal{C}_N$. Then $\rho(X,x) = \sum_{i=1}^N I(X_i \neq x_i)$. Since X is uniform on $\mathcal{C}_N$, each X_i is uniform on $\{0,1\}$, hence Bernoulli with parameter $1/2$. Likewise, for every $x_i \in \{0,1\}$, $I(X_i \neq x_i)$ is Bernoulli with parameter $1/2$. The coordinates X_i are independent, so the variables $\delta_i = I(X_i \neq x_i)$ are independent as well. Thus $\rho(X,x)$ is a sum of N independent Bernoulli random variables of parameter $1/2$: it is a binomial random variable of parameters N and $1/2$.

3. Let $\delta_1, \dots, \delta_N$ be independent Bernoulli random variables of parameter $1/2$. By 1), 2) and Chernoff's bound,

$$|\{y \in \mathcal{C}_N : \rho(y,x) \leq k\}| = 2^N \mathbb{P}\big[\rho(X,x) \leq k\big] = 2^N \mathbb{P}\Big[\sum_{i=1}^N \delta_i \leq k\Big]$$

$$= 2^N \mathbb{P}\Big[\frac{1}{N}\sum_{i=1}^N \frac{1}{2} - \delta_i \geq \frac{1}{2} - \frac{k}{N}\Big] \leq 2^N \exp\big(-N h_{1/2}(1/2 - k/N)\big)$$

where

$$h_{1/2}(1/2 - k/N) = \log 2 - \frac{k}{N}\log\Big(\frac{N}{k}\Big) - \Big(1 - \frac{k}{N}\Big)\log\Big(\frac{N}{N-k}\Big).$$

Therefore

$$|\{y \in \mathcal{C}_N : \rho(y,x) \leq k\}| \leq \Big(\frac{N-k}{k}\Big)^k \Big(\frac{N}{N-k}\Big)^N \leq \Big(\frac{eN}{k}\Big)^k.$$

4. Start with $x_1 = (0, \dots, 0) \in \mathcal{C}_N$ and set

$$\Omega_1 = \{y \in \mathcal{C}_N : \rho(y, x_1) \leq k\}.$$

Then $|\Omega_1| \leq \big(eN/k\big)^k$. Choose x_2 in $\mathcal{C}_N \setminus \Omega_1$. There are at least $2^N - \big(eN/k\big)^k \geq 1$ possible choices. Set

$$\Omega_2 = \{y \in \mathcal{C}_N : \rho(y, x_2) \leq k\}.$$

Choose x_3 in $\mathcal{C}_N \setminus (\Omega_1 \cup \Omega_2)$. There are at least $2^N - 2\big(eN/k\big)^k \geq 1$ choices, and x_3 is k-separated from x_1 and x_2. Iterate until step p satisfies

$$2^N - (p+1)\big(eN/k\big)^k < 1 \leq 2^N - p\big(eN/k\big)^k.$$

This yields the claimed cardinality.

2 Concentration

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Exercise 2.1 (Laplace transform of a Gaussian)
Let $g \sim \mathcal{N}(\theta, \sigma^2)$. For every $\lambda \in \mathbb{R}$,

$$\mathbb{E}\exp(\lambda g) = \exp\left(\lambda\theta + \frac{\lambda^2\sigma^2}{2}\right).$$

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Solution of the exercise 2.1 If $\eta \sim \mathcal{N}(0,1)$, then

$$\mathbb{E}\exp(\lambda g) = \mathbb{E}\exp(\lambda(\theta+\sigma\eta)) = \exp(\lambda\theta)\mathbb{E}\exp(\lambda\sigma\eta).$$

Moreover, for every $\mu \in \mathbb{R}$,

$$\mathbb{E}\exp(\mu\eta) = \int_{\mathbb{R}} \exp(\mu t)\exp(-t^2/2)\frac{dt}{\sqrt{2\pi}} = \int_{\mathbb{R}} \exp\left(-\frac{1}{2}(t^2-2\mu t)\right)\frac{dt}{\sqrt{2\pi}}$$
$$= \exp\left(\frac{\mu^2}{2}\right)\int_{\mathbb{R}} \exp\left(-\frac{(t-\mu)^2}{2}\right)\frac{dt}{\sqrt{2\pi}} = \exp\left(\frac{\mu^2}{2}\right).$$

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Exercise 2.2 (Bernoulli estimate)

Let $(A_i)_{i\geq 1}$ be a sequence of independent events such that $a := \sum_{i\geq 1}\mathbb{P}(A_i) < \infty$. For every n, the *Bernoulli estimate* states that

$$\mathbb{P}\Big[\sum_{i\geq 1} I_{A_i} \geq n\Big] \leq \frac{a^n}{n!} \leq \Big(\frac{ea}{n}\Big)^n.$$

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Solution of the exercise 2.2 By the union bound and independence,

$$\mathbb{P}\big[\sum_{i\geq 1} I_{A_i} \geq n\big] = \mathbb{P}\Big[\cup_{i_1<\ldots<i_n} A_{i_1}\cap\ldots\cap A_{i_n}\Big] \leq \sum_{i_1<\ldots<i_n}\prod_{j=1}^n \mathbb{P}(A_{i_j})$$
$$= \frac{1}{n!}\sum_{i_1\neq\cdots\neq i_n}\prod_{j=1}^n \mathbb{P}(A_{i_j}) \leq \frac{1}{n!}\sum_{(i_1,\cdots,i_n)}\prod_{j=1}^n \mathbb{P}(A_{i_j}) = \frac{1}{n!}\Big(\sum_i \mathbb{P}(A_i)\Big)^n = \frac{a^n}{n!}.$$

One concludes since $n! \geq (n/e)^n$.

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Exercise 2.3 (Chernoff's inequality)

Let $\delta_1,\ldots,\delta_n$ be i.i.d. Bernoulli random variables with mean δ (also called *selectors*), defined by $\mathbb{P}[\delta_1=1] = 1-\mathbb{P}[\delta_1=0] = \delta$. Chernoff's inequality states that, for every t with $-\delta < t < 1-\delta$, writing

$$h_\delta(t) := (1-\delta-t)\log\Big(\frac{1-\delta-t}{1-\delta}\Big) + (\delta+t)\log\Big(\frac{\delta+t}{\delta}\Big),$$

one has, for every $0 \leq t < 1-\delta$,

$$\mathbb{P}\Big[\frac{1}{n}\sum_{i=1}^n \delta_i - \delta \geq t\Big] \leq \exp(-nh_\delta(t)),$$

and, for every $0 \leq t < \delta$,

$$\mathbb{P}\Big[\frac{1}{n}\sum_{i=1}^n \delta_i - \delta \leq -t\Big] \leq \exp(-nh_\delta(-t)).$$

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Solution of the exercise 2.3 Use the Cramér–Chernoff method: for every $t > 0$,

$$\mathbb{P}\Big[\frac{1}{n}\sum_{i=1}^{n}\delta_i - \delta \geq t\Big] \leq \exp\big(-\psi^*_{\bar{\delta}_n}(t)\big),$$

where $\psi^*_{\bar{\delta}_n}$ is the convex conjugate of the Laplace transform of $\bar{\delta}_n := n^{-1}\sum_{i=1}^n \delta_i - \delta$, defined by $\psi_{\bar{\delta}_n}(\lambda) := \log \mathbb{E}\exp(\lambda\bar{\delta}_n)$ for every $\lambda > 0$ and

$$\psi^*_{\bar{\delta}_n}(t) = \sup_{\lambda>0}\big[\lambda t - \psi_{\bar{\delta}_n}(\lambda)\big].$$

By independence, $\psi_{\bar{\delta}_n}(\lambda) = n\psi_{\delta_1-\delta}(\lambda/n)$, hence $\psi^*_{\bar{\delta}_n}(t) = n\psi^*_{\delta_1-\delta}(t)$. A direct computation gives, for every $\lambda \geq 0$,

$$\psi_{\delta_1-\delta}(\lambda) = \log\big(\delta\exp(\lambda) + 1 - \delta\big) - \lambda\delta$$

and, for every $0 \leq t < 1 - \delta$,

$$\psi^*_{\delta_1-\delta}(t) = (1-\delta-t)\log\Big(\frac{1-\delta-t}{1-\delta}\Big) + (\delta+t)\log\Big(\frac{\delta+t}{\delta}\Big).$$

This proves the first inequality. The second one follows in the same way applied to $1 - \delta_i$, which replaces $h_\delta(t)$ by $h_\delta(-t)$.

****

Exercise 2.4 (Bounded random variables are sub-Gaussian)
Let X be a centered real random variable such that $a \leq X \leq b$ almost surely. Then for every $\lambda > 0$,

$$\mathbb{E}[e^{\lambda X}] \leq \exp\Big(\frac{\lambda^2(b-a)^2}{8}\Big).$$

****

Solution of the exercise 2.4 Let $\lambda > 0$. By convexity of the exponential, for every $x \in [a, b]$,

$$\exp(\lambda x) = \exp\Big[\Big(\frac{x-a}{b-a}\Big)\lambda b + \Big(\frac{b-x}{b-a}\Big)\lambda a\Big] \leq \Big(\frac{x-a}{b-a}\Big)\exp(\lambda b) + \Big(\frac{b-x}{b-a}\Big)\exp(\lambda a).$$

With $p = -a/(b-a)$ and $u = \lambda(b-a)$,

$$\begin{aligned}
&\mathbb{E}\exp(\lambda X) \leq \Big(\frac{-a}{b-a}\Big)\exp(\lambda b) + \Big(\frac{b}{b-a}\Big)\exp(\lambda a)\\
&\leq \exp(\lambda a)\left(\frac{b}{b-a} - \frac{a}{b-a}\exp\left(\lambda(b-a)\right)\right)\\
&\leq \exp\left[-\underbrace{\Big(\frac{-a}{b-a}\Big)}_{=p}\underbrace{\lambda(b-a)}_{=u}\right]\left(1 + \underbrace{\frac{a}{b-a}}_{=-p} + \underbrace{\frac{-a}{b-a}}_{=p}\exp\left(\underbrace{\lambda(b-a)}_{=u}\right)\right)\\
&= \exp(-pu)\left(1 - p + p\exp(u)\right) := \exp(\phi(u))
\end{aligned}$$

where $\phi(u) = -pu + \log(1 - p + pe^u)$.

Recall that for all $\alpha, \beta > 0$, one has $\alpha\beta \leq (\alpha + \beta)^2/4$. Thus, for every $u > 0$,

$$\phi'(u) = -p + \frac{pe^u}{1 - p + pe^u} \text{ and } \phi''(u) = \frac{p(1-p)e^u}{(1 - p + pe^u)^2} \leq \frac{1}{4}.$$

A Taylor expansion of order 2 gives $\phi(u) \leq \phi(0) + u\phi'(0) + u^2/8 = \lambda^2(b-a)^2/8$.

****

Exercise 2.5 (Hoeffding's inequality)

Let $X_1, \ldots, X_n$ be independent centered real random variables such that for every $i = 1, \ldots, n$, $a_i \leq X_i \leq b_i$. For every $t > 0$,

$$\mathbb{P}\Big[\frac{1}{n}\sum_{i=1}^n X_i > t\Big] \leq \exp\Big(-\frac{2nt^2}{\frac{1}{n}\sum_{i=1}^n (b_i - a_i)^2}\Big).$$

****

Solution of the exercise 2.5 Use the Cramér–Chernoff method:

$$\mathbb{P}\Big[\frac{1}{n}\sum_{i=1}^n X_i > t\Big] \leq \inf_{\lambda > 0} e^{-\lambda t}\mathbb{E}\exp\Big(\frac{\lambda}{n}\sum_{i=1}^n X_i\Big) = \inf_{\lambda > 0} e^{-\lambda t}\prod_{i=1}^n \mathbb{E}\exp(\lambda X_i/n).$$

Since $a_i \leq X_i \leq b_i$, the previous exercise gives

$$\mathbb{E}\exp(\lambda X_i/n) \leq \exp\Big(\frac{\lambda^2(b_i - a_i)^2}{8n^2}\Big).$$

Therefore

$$\mathbb{P}\Big[\frac{1}{n}\sum_{i=1}^n X_i > t\Big] \leq \inf_{\lambda > 0} \exp\Big(-\lambda t + \frac{\lambda^2}{8n^2}\sum_{i=1}^n (b_i - a_i)^2\Big) = \exp\Big(-\frac{2nt^2}{\frac{1}{n}\sum_{i=1}^n (b_i - a_i)^2}\Big).$$

****

Exercise 2.6 (Bernstein's inequality)

Let $X_1, \ldots, X_n$ be independent random variables and let $M > 0$ be such that for every $i = 1, \ldots, n$, $\mathbb{E}X_i = 0$ and $|X_i| \leq M$ almost surely. Set $\sigma^2 = n^{-1}\sum_{i=1}^n \mathbb{E}X_i^2$. For every $t > 0$,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\frac{n\sigma^2}{M^2}h\left(\frac{Mt}{\sigma^2}\right)\right) \leq \exp\left(-\frac{3n}{8}\min\left(\frac{t^2}{\sigma^2}, \frac{t}{M}\right)\right),$$

where $h(u) = (1+u)\log(1+u) - u$ for every $u > 0$.

****

Solution of the exercise 2.6 Let $t > 0$. By Markov's inequality and independence,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n X_i \geq t\right) \leq \inf_{\lambda > 0} \exp(-\lambda t)\mathbb{E}\exp\left(\frac{\lambda}{n}\sum_{i=1}^n X_i\right) = \inf_{\lambda > 0} \exp(-\lambda t)\prod_{i=1}^n \mathbb{E}\exp\left(\frac{\lambda X_i}{n}\right). \quad (3)$$

Since $\mathbb{E}X_i = 0$ and $|X_i| \leq M$ for every i,

$$\mathbb{E}\exp\left(\frac{\lambda X_i}{n}\right) = 1 + \sum_{k\geq 2}\frac{\lambda^k \mathbb{E}X_i^k}{n^k k!} \leq 1 + \mathbb{E}X_i^2 \sum_{k\geq 2}\frac{\lambda^k M^{k-2}}{n^k k!} = 1 + \frac{\mathbb{E}X_i^2}{M^2}\left(\exp\left(\frac{\lambda M}{n}\right) - \left(\frac{\lambda M}{n}\right) - 1\right).$$

Since $1 + u \leq \exp(u)$ for every $u \in \mathbb{R}$,

$$\prod_{i=1}^n \mathbb{E}\exp\left(\frac{\lambda X_i}{n}\right) \leq \exp\left(\frac{\sum_{i=1}^n \mathbb{E}X_i^2}{M^2}\left(\exp\left(\frac{\lambda M}{n}\right) - \left(\frac{\lambda M}{n}\right) - 1\right)\right).$$

By definition of σ and (3), for every $t > 0$,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n X_i \geq t\right) \leq \inf_{\lambda>0}\exp\left(\frac{n\sigma^2}{M^2}\left(\exp\left(\frac{\lambda M}{n}\right) - \left(\frac{\lambda M}{n}\right) - 1\right) - \lambda t\right).$$

The result follows by taking λ such that $(1 + tM/\sigma^2) = \exp(\lambda M/n)$.

By Taylor's formula, $h(u) \geq u^2/(2 + 2u/3)$ for every $u > 0$. Hence $h(u) \geq 3u/8$ if $u \geq 1$, and $h(u) \geq 3u^2/8$ if $u \leq 1$.

****

Exercise 2.7 (Counting and concentration)
Prove that for every N and every $1 \leq D \leq N$,

$$\sum_{i=0}^D \binom{N}{i} \leq \left(\frac{eN}{D}\right)^D.$$

One may consider N Bernoulli random variables $\delta_1, \ldots, \delta_N$ of mean $1/2$ and use a concentration inequality to bound $\mathbb{P}[\sum_{i=1}^N \delta_i \leq D]$.

****

Solution of the exercise 2.7 The map $D \mapsto \left(eN/D\right)^D$ is increasing. Thus, when $D \geq N/2$, one has $\left(eN/D\right)^D \geq (\sqrt{2e})^N > 2^N$, so the inequality holds in this case. One may now assume $D < N/2$.

Let $\delta_1, \ldots, \delta_N$ be independent Bernoulli random variables of mean $1/2$, and set $S = \sum_{i=1}^N \delta_i$. Then

$$\frac{1}{2^N}\sum_{i=0}^D \binom{N}{i} = \mathbb{P}\big[S \leq D\big].$$

Since $\mathbb{E}S = N/2$ and $N/2 - D > 0$,

$$\mathbb{P}\big[S \leq D\big] = \mathbb{P}\Big[\sum_{i=1}^N \mathbb{E}\delta_i - \delta_i \geq \frac{N}{2} - D\Big] = \mathbb{P}\Big[\frac{1}{N}\sum_{i=1}^N X_i \geq t\Big]$$

where $X_i = \mathbb{E}\delta_i - \delta_i$ and $t = 1/2 - D/N > 0$. This is a concentration problem for an empirical mean. Hoeffding's inequality applies: $X_i \in [-1/2, 1/2]$ a.s., so $b_i - a_i = 1$ and

$$\mathbb{P}\Big[\frac{1}{N}\sum_{i=1}^N X_i \geq t\Big] \leq \exp\big(-2Nt^2\big).$$

Bernstein's inequality gives, since $|X_i| \le 1/2$ a.s. and $\mathrm{var}(X_i) = 1/4$,

$$\mathbb{P}\Big[\frac{1}{N}\sum_{i=1}^{N} X_i \ge t\Big] \le \exp\big(-c_0 N \min(t^2, t)\big).$$

In this regime Hoeffding's inequality is the sharper of the two: controlling the variance of the X_i adds nothing beyond the fact that they are bounded.

One may also apply Chernoff's inequality: for independent Bernoulli random variables of parameter δ,

$$\mathbb{P}\Big[\frac{1}{N}\sum_{i=1}^{N} \delta - \delta_i \ge t\Big] \le \exp\Big(-N h_\delta(t)\Big)$$

where

$$h_\delta(t) = (1-\delta-t)\log\Big(\frac{1-\delta-t}{1-\delta}\Big) + (\delta+t)\log\Big(\frac{\delta+t}{\delta}\Big).$$

The Bernoulli estimate then yields

$$\begin{aligned}\sum_{i=0}^{D}\binom{N}{i} &= 2^N \mathbb{P}\big[S \le D\big] \le 2^N \exp\big(-N h_{1/2}(1/2 - D/N)\big)\\ &= \exp\Big[D\log\Big(\frac{N}{D}\Big) + (N-D)\log\Big(\frac{N}{N-D}\Big)\Big] \le \Big(\frac{eN}{D}\Big)^D\end{aligned}$$

since $\log(1+x) \le x$ for every $x \ge 0$, and therefore

$$(N-D)\log\Big(\frac{N}{N-D}\Big) = (N-D)\log\Big(1 + \frac{D}{N-D}\Big) \le D.$$

****

Exercise 2.8 (Johnson–Lindenstrauss lemma)

This famous high-dimensional geometry result from the 1980s has recently become central to dimension-reduction methods: given p points $x_1, \dots, x_p$ in $\mathbb{R}^N$ (thought of as high-dimensional), there exists a matrix $A : \mathbb{R}^N \to \mathbb{R}^k$ with $k \sim \log p$ such that for all $i, j = 1, \dots, p$, $\|A(x_i - x_j)\|_2 \sim \|x_i - x_j\|_2$.

In other words, one can map p points from a high-dimensional space into a space of dimension only $\log p$, while approximately preserving the original Euclidean distances in $\mathbb{R}^N$. This is a dimension-reduction result: one passes from the "large" dimension N to a low-dimensional space $\mathbb{R}^k$ while keeping the mutual distances of the p points.

More precisely: *there exists an absolute constant $c_0 > 0$ such that for every $0 < \varepsilon < 1$, every set of p points $x_1, \dots, x_p$ in $\mathbb{R}^N$, and every $k \ge c_0(\log p)/\varepsilon^2$, there exists a linear map $A : \mathbb{R}^N \to \mathbb{R}^k$ such that for all $i, j = 1, \dots, p$,*

$$\sqrt{1-\varepsilon}\,\|x_i - x_j\|_2 \le \|A(x_i - x_j)\|_2 \le \sqrt{1+\varepsilon}\,\|x_i - x_j\|_2 .$$

****

Solution of the exercise 2.8 The argument is probabilistic and relies on concentration of Rademacher random variables (Gaussian variables also work, and the proof is then simpler; the point here is

to use Hoeffding's inequality). Take A to be the random matrix with i.i.d. Rademacher entries $(\varepsilon_{ij} : 1 \le i \le k,\ 1 \le j \le N)$, normalized by $k^{-1/2}$:

$$A = \frac{1}{\sqrt{k}}\big(\varepsilon_{ij}\big)_{\substack{1\le i\le k\\ 1\le j\le N}} = \frac{1}{\sqrt{k}}\begin{pmatrix} \boldsymbol{\varepsilon}_1^\top \\ \vdots \\ \boldsymbol{\varepsilon}_k^\top \end{pmatrix} : \mathbb{R}^N \to \mathbb{R}^k,$$

where $\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_k$ are i.i.d. Rademacher vectors in $\mathbb{R}^N$.

First, for every $x \in \mathbb{R}^N$, $\mathbb{E}\langle \boldsymbol{\varepsilon}, x\rangle^2 = \|x\|_2^2$. Indeed,

$$\mathbb{E}\langle \boldsymbol{\varepsilon}, x\rangle^2 = \mathbb{E}\Big(\sum_{j=1}^N \varepsilon_j x_j\Big)^2 = \sum_{j_1,j_2=1}^N \mathbb{E}\varepsilon_{j_1}\varepsilon_{j_2}x_{j_1}x_{j_2} = \sum_{j=1}^N x_j^2 = \|x\|_2^2 .$$

The random vector $\boldsymbol{\varepsilon}$ is isotropic.

Fix $r, s \in \{1, \dots, p\}$ and consider the Euclidean norm of $A(x_r - x_s)$. Then

$$\mathbb{E}\,\|A(x_r - x_s)\|_2^2 = \mathbb{E}\frac{1}{k}\sum_{i=1}^k \langle \boldsymbol{\varepsilon}_i, x_r - x_s\rangle^2 = \|x_r - x_s\|_2^2 .$$

Hence, for every $t > 0$,

$$\mathbb{P}\Big[\big|\, \|A(x_r - x_s)\|_2^2 - \|x_r - x_s\|_2^2 \,\big| \ge t\Big] = \mathbb{P}\Big[\Big|\frac{1}{k}\sum_{i=1}^k X_i - \mathbb{E}X_i\Big| \ge t\Big],$$

where $X_i = \langle \boldsymbol{\varepsilon}_i, x_r - x_s\rangle^2$. In Orlicz norms, $\|X_i\|_{\psi_1} = \big\|\langle \boldsymbol{\varepsilon}_i, x_r - x_s\rangle^2\big\|_{\psi_1} = \big\|\langle \boldsymbol{\varepsilon}_i, x_r - x_s\rangle\big\|_{\psi_2}^2$. Hoeffding's inequality gives

$$\big\|\langle \boldsymbol{\varepsilon}_i, x_r - x_s\rangle\big\|_{\psi_2} \le c_0 \|x_r - x_s\|_2 .$$

Therefore, for every $0 < \varepsilon < 1$,

$$\mathbb{P}\Big[\big|\, \|A(x_r - x_s)\|_2^2 - \|x_r - x_s\|_2^2 \,\big| \ge \varepsilon \|x_r - x_s\|_2^2\Big] \le 2\exp\big(-ck\varepsilon^2\big).$$

This estimate concerns a single pair (x_r, x_s). There are at most p^2 such pairs. The union bound yields

$$\mathbb{P}\Big[\exists\, 1 \le s, r \le p : \big|\, \|A(x_r - x_s)\|_2^2 - \|x_r - x_s\|_2^2 \,\big| \ge \varepsilon \|x_r - x_s\|_2^2\Big] \le 2p^2\exp\big(-ck\varepsilon^2\big).$$

In particular, when $k \ge (2\log p + \log 4)/(c\varepsilon^2)$, with probability at least $1/2$ one has, for all $1 \le s, r \le p$,

$$(1-\varepsilon)\|x_s - x_r\|_2^2 \le \|A(x_r - x_s)\|_2^2 \le (1+\varepsilon)\|x_r - x_s\|_2^2 .$$

****

Exercise 2.9 (The coupon collector problem)

An urn contains n distinct coupons. At each draw, a coupon is chosen uniformly at random and then replaced. How many draws are needed in order to be reasonably sure (that is, with probability at least $1/2$) of having seen every coupon at least once? The answer is of order $\mathcal{O}(n \log n)$.

Recall that the geometric distribution of parameter $0 < \delta < 1$ is the law of the trial N of the first success in a sequence of independent Bernoulli trials of parameter δ. It may be defined by

$$N = \min(n : \delta_n = 1),$$

where $(\delta_n)_n$ is a sequence of i.i.d. Bernoulli random variables of parameter δ.

1 Determine the law, the expectation, the variance and the tail of N: $\mathbb{P}[N > k]$ for every $k \in \mathbb{N}^*$.

Let $i \in \{1, \dots, n\}$, and suppose that $i-1$ distinct coupons have just been observed. Let T_i be the number of additional draws needed to observe a new coupon, that is, the first time an i-th coupon, different from the $i-1$ already seen, appears.

2 What is the law of T_i? What are the means of T_1 and T_n? Interpret these values.

3 Let T be the first time at which every coupon in the urn has been observed. Express T in terms of the T_i, and compute its mean.

4 Compute the variance of T and, using Chebyshev's inequality, bound the probability that T deviates from its mean.

****

Solution of the exercise 2.9

1. For every $k \in \mathbb{N}^*$,
$$\mathbb{P}\big[N = k\big] = (1-\delta)^{k-1}\delta.$$
In particular, $\mathbb{E}N = \delta^{-1}$, $\mathrm{var}(N) = (1-\delta)\delta^{-2}$, and for every $k \in \mathbb{N}^*$,
$$\mathbb{P}\big[N > k\big] = 1 - \sum_{i=1}^{k}(1-\delta)^{i-1}\delta = (1-\delta)^k = \exp\Big(-k\log\Big(\frac{1}{1-\delta}\Big)\Big).$$
Thus N is a sub-exponential random variable.

2. The probability of observing a new coupon at this stage is $(n-i+1)/n$. Hence T_i is geometric with parameter
$$\delta_i = 1 - \frac{i-1}{n}.$$
In particular $\delta_1 = 1$: the first draw is certainly a new coupon. Also $\delta_n = 1/n$, so once $n-1$ distinct coupons have been observed, one needs on average n further draws just to see the last coupon (since $\mathbb{E}T_n = \delta_n^{-1} = n$).

3. The first time at which all n coupons have been observed at least once is $T = T_1 + T_2 + \ldots + T_n$: a sum of independent geometric random variables with parameters $\delta_1 = 1, \delta_2 = 1 - 1/n, \cdots, \delta_n = 1/n$.

The expected number of draws needed to see every coupon at least once is therefore

$$\mathbb{E}T = \sum_{i=1}^{n} \mathbb{E}T_i = \sum_{i=1}^{n} \delta_i^{-1} = \frac{n}{n} + \frac{n}{n-1} + \cdots + \frac{n}{1} = n \sum_{i=1}^{n} \frac{1}{i} = nH_n,$$

where $H_n = \sum_{i=1}^{n} i^{-1}$ is the harmonic number. As $n \to \infty$,

$$H_n = \log n + \gamma + \frac{1}{2n} + o(1/n),$$

where $\gamma \sim 0.577$ is Euler's constant. Hence

$$\mathbb{E}T = n \log n + \gamma n + \frac{1}{2} + o(1).$$

On average, the first time at which every coupon has been observed is of order $n \log n$.

4. The variance of T is

$$\mathrm{var}(T) = \sum_{i=1}^{n} \mathrm{var}(T_i) = \sum_{i=1}^{n} \frac{n(i-1)}{(n-i+1)^2} \leq n^2 \sum_{i=1}^{n} \frac{1}{i^2} \leq 2n^2.$$

Chebyshev's inequality gives, for every $t > 0$,

$$\mathbb{P}[|T - \mathbb{E}T| \geq tn] \leq \frac{\mathrm{var}(T)}{(tn)^2} \leq \frac{2}{t^2}.$$

Thus, with probability greater than $1 - 2t^{-2}$, one has $T \leq n \log n + (\gamma + t)n + 1/2 + o(1)$.

****

Exercise 2.10 (Bernstein's inequality with the empirical variance)

Part one

Let V, $V_1, \ldots, V_n$ be i.i.d. random variables with nonpositive values. Let $\bar{V} = \frac{1}{n} \sum_{i=1}^{n} V_i$.

1. **Show that $\exp(u) \leq 1 + u + \frac{u^2}{2}$ for $u \leq 0$.**
2. **Show that $\log \mathbb{E} \exp(sV) \leq s\mathbb{E}V + \frac{s^2}{2}\mathbb{E}(V^2)$ for every $s \geq 0$.**
3. **Deduce, by Chernoff's argument, that for every $\lambda > 0$,**

$$\mathbb{P}\big(\bar{V} - \mathbb{E}V > t\big) \leq \exp\bigg(- \lambda t + \frac{\lambda^2 \mathbb{E}(V^2)}{2n} \bigg).$$

4. **Deduce that with probability at least $1 - \varepsilon$,**

$$\bar{V} - \mathbb{E}V \leq \sqrt{\frac{2\mathbb{E}(V^2) \log(1/\varepsilon)}{n}}.$$

Part two

Let X, $X_1, \ldots, X_n$ be i.i.d. random variables with values in $[0, 1]$. Let $\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$, let $\sigma = \sqrt{\mathbb{E}[(X - \mathbb{E}X)^2]}$ be the standard deviation of X, let $\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2}$ be the empirical standard deviation, let $\varepsilon > 0$ and $L = \frac{\log(1/\varepsilon)}{n}$.

5. Using the previous question, show that with probability at least $1-\varepsilon$,

$$\sigma^2 \leq \frac{1}{n}\sum_{i=1}^{n}(X_i - \mathbb{E}X)^2 + \sigma\sqrt{2L}.$$

6. Show that with probability at least $1-2\varepsilon$,

$$\left|\bar{X} - \mathbb{E}X\right| \leq \sigma\sqrt{2L} + \frac{L}{3} \tag{4}$$

and that with probability at least $1-3\varepsilon$, (4) and

$$\sigma^2 \leq \hat{\sigma}^2 + \left|\bar{X} - \mathbb{E}X\right|^2 + \sigma\sqrt{2L} \tag{5}$$

hold simultaneously.

7. The goal of this question is to show that whenever (4) and (5) hold,

$$\sigma \leq \hat{\sigma} + 1.8\sqrt{L}. \tag{6}$$

7.a Show that (6) holds when $1.8\sqrt{L} \geq \frac{1}{2}$.

7.b When $1.8\sqrt{L} < \frac{1}{2}$, show that

$$\sigma^2 \leq \hat{\sigma}^2 + \frac{\sqrt{L}\sigma}{3.6} + \frac{2}{3\times(3.6)^2}\sigma\sqrt{2L} + \frac{L}{9\times(3.6)^2} + \sigma\sqrt{2L},$$

and deduce $\sigma \leq \frac{1.77}{2}\sqrt{L} + \sqrt{\hat{\sigma}^2 + 0.8L}$.

7.c Conclude.

8 Deduce from the previous questions that with probability at least $1-\varepsilon$,

$$\left|\bar{X} - \mathbb{E}X\right| \leq \hat{\sigma}\sqrt{\frac{2\log(3\varepsilon^{-1})}{n}} + \frac{3\log(3\varepsilon^{-1})}{n},$$

and compare this with (4).

****

Solution of the exercise 2.10

1. Study the function $u \mapsto \exp(u) - 1 - u - \frac{u^2}{2}$.
2. Apply question 1), together with $\log(1+u) \leq u$.
3. $\mathbb{P}(\bar{V} - \mathbb{E}V > t) = \mathbb{P}\big(e^{\lambda(\bar{V}-\mathbb{E}V)} > e^{\lambda t}\big) \leq e^{-\lambda(t+\mathbb{E}V)}\mathbb{E}[e^{\lambda\bar{V}}] \leq e^{-\lambda t}e^{\frac{\lambda^2\mathbb{E}(V^2)}{2n}}$, by question 2) and the i.i.d. assumption on $V_1, \ldots, V_n$.
4. If $\mathbb{E}(V^2) = 0$ the claim is trivial. Otherwise optimize in λ (take $\lambda = nt/\mathbb{E}V^2$) and choose t so that $e^{-nt^2/(2\mathbb{E}V^2)} = \varepsilon$.
5. Apply 4) to $V = -(X - \mathbb{E}X)^2$, using $\mathbb{E}V^2 \leq \sigma^2$.

6. Write the classical Bernstein inequality for X and for $-X$, and take the union bound over the two resulting inequalities.

   The second inequality is question 5) after expanding and simplifying the sum. By the union bound, since question 5) and inequality (1) hold with probability at least $1-\varepsilon$ and $1-2\varepsilon$ respectively, (1) and (2) hold simultaneously on an event of probability at least $1-3\varepsilon$.

7.a The variance of a random variable with values in $[0,1]$ is at most $1/4$, so its standard deviation is at most $1/2$.

7.b Insert (1) into (2), and use $\sqrt{L} < \frac{1}{3.6}$ and $\sigma \leq 1/2$ where needed. For the second part, reduce to the quadratic inequality $\sigma^2 - 1.77\sqrt{L}\sigma - (\hat{\sigma}^2 + \frac{L}{100}) \leq 0$.

7.c Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ for $a, b \geq 0$, one obtains (3) from $1.77/2 + \sqrt{0.8} < 1.8$.

7. Combine (1) and (3), and conclude from $1.8\sqrt{2} + 1/3 < 3$. At the price of a larger constant (3 instead of $1/3$) and a slightly different confidence level, (1) remains valid when the variance is replaced by its empirical counterpart: a Bernstein inequality with the empirical variance. This is useful for building observable, nonasymptotic confidence intervals that are sharper than those coming from Hoeffding's bound.

3 Compressed sensing

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Exercise 3.1 (Incoherent measurements via the twelve-coin problem)

The goal of this exercise is to become familiar with measurements, sparsity, dictionaries and incoherence, through the twelve-coin problem. For more details, see

a web page on the twelve-coin problem.

The twelve-coin problem is the following. One is given twelve coins, identical in size, shape, and so on, except that a single one among them has a mass different from the other eleven. One does not know whether this coin is heavier or lighter than the others; one only knows that exactly one coin has a different mass. A two-pan balance is available. How can one find this coin among the twelve with at most 3 weighings?

1. **Model this problem in the compressed-sensing framework, and point out the differences with the classical CS setting.**
2. **Construct the "trivial" measurements that solve the problem in 7 weighings.**
3. **Define the notion of sparsity and the associated dictionary for this problem.**
4. **Propose 3 (nonadaptive) measurement vectors that solve the problem.**

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Solution of the exercise 3.1

1. The problem can be modelled in the CS framework as follows. Let $x \in \mathbb{R}^{12}$ be the vector of the weights of the 12 coins. The "signal" x has the form $x_i = p_0$ for every $i \in \{1, \dots, 12\} \setminus \{i_1\}$, except for one index i_1 for which $x_{i_1} = p_1$, with $p_0 \neq p_1$. The i-th weighing is associated with a measurement vector $X_i \in \mathbb{R}^{12}$, where

$$(X_i)_j = \begin{cases} 1 & \text{if coin } j \text{ is placed on the left pan,} \\ -1 & \text{if coin } j \text{ is placed on the right pan,} \\ 0 & \text{if coin } j \text{ is not used in weighing } i. \end{cases}$$

   Here one only observes the sign of the measurements $\langle x, X_i \rangle$, that is

$$\operatorname{sign}(\langle x, X_i \rangle) = \begin{cases} 1 & \text{when the balance tips to the left,} \\ -1 & \text{when the balance tips to the right,} \\ 0 & \text{when the balance stays level.} \end{cases}$$

   Unlike the classical CS setting, one only observes the sign of $\langle x, X_i \rangle$. Moreover, the goal is not to reconstruct x, but only to locate the coordinate at which x differs from the others, that is, to reconstruct its support.

   These differences with the classical CS problem, as introduced in the previous chapters, have been studied in recent years. Observing the sign of the measurements is a form of quantization called *one-bit compressed sensing*; recovering only the support of x is called *support recovery*. These two variants of CS are not presented in the course. In all of these problems, however, the central role of the incoherence between the measurements and the basis (respectively, the dictionary) in which the sparsity of the signal of interest is expressed remains.

   Furthermore, the classical statement of the twelve-coin problem allows **adaptive** measurements, for which one may adapt the experimental design (that is, choose the measurement vectors) according to the previous outcomes. This is not the classical CS framework, which only considers nonadaptive measurements (whose measurement vectors do not depend on previous outcomes). In fact there is a nonadaptive solution of the twelve-coin problem: the same 3 measurements can be used in every case.

2. The trivial strategy with 7 weighings compares the coins in pairs, one on each pan. In the worst case, 7 measurements are needed: if the coin of different mass is in the last pair, 6 weighings identify the pair that contains it, and one last weighing determines which of the two coins differs from the others. In this strategy the first 6 measurement vectors are

$$(1, -1, 0, \dots, 0)^\top, (0, 0, 1, -1, 0, \dots, 0)^\top, \cdots, (0, \dots, 0, 1, -1)^\top,$$

   and the seventh is built from the weighing that identified which pair contains the different coin.

3. The previous strategy uses far too many measurements, because the measurement vectors are not incoherent with the "basis" in which the signal x is sparse. One must first identify the low-dimensional structure of the problem. Here the sparse vector is the gradient of x:

$$\nabla x = (x_2 - x_1, \dots, x_{12} - x_{11})^\top \in \mathbb{R}^{11}.$$

Indeed, $\|\nabla x\|_0 \leq 2$.

Let us translate this sparsity of ∇x into a sparsity of x for a suitable dictionary (the next goal is to build vectors that are incoherent with the elements of this dictionary). The discrete differentiation operator ∇ may be written in matrix form:

$$\nabla = \begin{pmatrix} -1 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & & & & \cdots & \vdots & \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1 \end{pmatrix} : \mathbb{R}^{12} \mapsto \mathbb{R}^{11}, \text{ and for } f_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^{12}$$

one obtains the discrete integration formula

$$x = \nabla^{-1}\nabla x + x_1 f_0, \tag{7}$$

where $\nabla^{-1} : \mathbb{R}^{11} \mapsto \mathbb{R}^{12}$ is the integration operator, given in matrix form by

$$\nabla^{-1} = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 1 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & & & & \cdots & \vdots & \\ 1 & 1 & 1 & 1 & \cdots & 1 & 1 \end{pmatrix} \in \mathbb{R}^{12\times 11}.$$

Along the way, one can draw the parallel with differentiation and integration of a function of a continuous variable: ∇^{-1} is discrete integration, ∇ is discrete differentiation, and f_0 plays the role of the constant function equal to 1. Thus (7) is the discrete form of

$$f(v) = \int_0^v f'(t)\,dt + f(0), \text{ for every } v > 0.$$

The dictionary in which x is s-sparse is given by the columns of ∇^{-1} (the component along f_0 is always nonzero). Define the dictionary $\{f_1, \ldots, f_{11}\}$ formed by the columns of ∇^{-1}:

$$f_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, f_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \cdots, f_{11} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Indeed,

$$x = (\nabla x)_1 f_1 + \cdots + (\nabla x)_{11} f_{11} + x_1 f_0,$$

where the coefficients $(\nabla x)_i$, $i = 1, \ldots, 11$, are the coordinates of the gradient ∇x, and are therefore s-sparse with $s \in \{1, 2\}$.

4. A nonadaptive solution is given by

$$X_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 1 \\ 0 \\ -1 \\ 1 \\ 0 \\ -1 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ -1 \\ -1 \\ -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} \text{ and } X_3 = \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}.$$

For each weighing, four coins are placed on the left pan and four on the right pan. Four coins are left aside at each measurement.

These measurements can be checked on a few examples.

If coin 1 is heavier: the outcomes are

$$\text{weighing } 1:1 \quad \text{weighing } 2:1 \quad \text{weighing } 3:-1.$$

From these weighings one deduces that coin 1 is indeed heavier than the others.

If coin 11 is lighter: the outcomes are

$$\text{weighing } 1:0 \quad \text{weighing } 2:-1 \quad \text{weighing } 3:-1.$$

From these weighings one deduces that coin 11 is indeed lighter.

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Exercise 3.2 (Discrete uncertainty principle and the Dirac comb)

The Fourier matrix (discrete Fourier transform) is given by

$$\Gamma : \begin{cases} \mathbb{C}^N & \to & \mathbb{C}^N \\ x & \to & \Gamma x = \hat{x} \end{cases} \quad \textbf{where } \Gamma = \Big(\frac{w^{(p-1)(q-1)}}{\sqrt{N}} \Big)_{1 \le p,q \le N} \textbf{ and } w = \exp(-2\mathbf{i}\pi/N).$$

Denote by $\bar{\Gamma}_1, \dots, \bar{\Gamma}_N$ the rows of Γ. Thus $\hat{x}_i = \langle \Gamma_i, x \rangle$ for every $i = 1, \dots, N$.

The discrete uncertainty principle says that if $x \in \mathbb{C}^N$ is nonzero, then

$$|\mathrm{supp}(x)| \times |\mathrm{supp}(\hat{x})| \ge N.$$

In particular, if x has small support, then its Fourier transform $\hat{x}$ necessarily has large support. This is the classical duality between sparse vectors and a "well spread" Fourier transform.

The goal is to prove this uncertainty principle and to show that it is sharp for *Dirac comb* signals.

1. **Write the discrete Plancherel identity.**
2. **Show that** $\|x\|_\infty \leq \frac{1}{\sqrt{N}} \|\hat{x}\|_1$**.**
3. **Deduce the result from the inequalities**

$$\|x\|_2^2 \leq |\mathrm{supp}(x)| \, \|x\|_\infty^2 \text{ and } \|\hat{x}\|_1 \leq \sqrt{|\mathrm{supp}(\hat{x})|} \, \|\hat{x}\|_2 .$$

4. **Assume that** $N = k^2$ **for some integer** k**. Define the *Dirac comb* x as the indicator of the set** $\{0, k, 2k, \cdots, k^2 - k\}$ **(coordinates are indexed by** $\{0, 1, \ldots, N-1\}$**).**

   4.a Show that $x = \hat{x}$ **for the Dirac comb.**

   4.b Show that $|\mathrm{supp}(x)| \times |\mathrm{supp}(\hat{x})| = N$ **for the Dirac comb.**

****

Solution of the exercise 3.2

1. Plancherel's identity reads $\|x\|_2^2 = \|\hat{x}\|_2^2$. Equivalently, the Fourier transform Γ is an isometry.
2. $\|x\|_\infty = \max_{1 \leq j \leq N} |\langle e_j, x \rangle|$, where $(e_1, \ldots, e_N)$ is the canonical basis of $\mathbb{C}^N$. Since Γ is an isometry,

$$\|x\|_\infty = \max_{1 \leq j \leq N} |\langle e_j, x \rangle| = \max_{1 \leq j \leq N} |\langle \Gamma e_j, \Gamma x \rangle| \leq \max_{1 \leq j \leq N} \|\Gamma e_j\|_\infty \, \|\Gamma x\|_1 = \frac{1}{\sqrt{N}} \|\hat{x}\|_1 .$$

3. Write I_x (resp. $I_{\hat{x}}$) for the support of x (resp. $\hat{x}$). Then

$$\|x\|_2^2 \leq |I_x| \, \|x\|_\infty^2 \leq \frac{|I_x| \, \|\hat{x}\|_1^2}{N} \leq \frac{|I_x||I_{\hat{x}}| \, \|\hat{x}\|_2^2}{N} = \frac{|I_x||I_{\hat{x}}| \, \|x\|_2^2}{N}.$$

   The result follows when $x \neq 0$.

4.a For $j \in \{0, 1, \ldots, N-1\}$,

$$\hat{x}_j = \frac{1}{\sqrt{N}} \sum_{r=0}^{N-1} w^{jr} x_r = \frac{1}{\sqrt{N}} \sum_{r=0}^{k-1} \exp(-2\mathbf{i}\pi jrk/k^2) = \frac{1}{k} \sum_{r=0}^{k-1} \exp(-2\mathbf{i}\pi jr/k)$$
$$= \begin{cases} 1 & \text{if } j \in \{0, k, 2k, \cdots, k^2 - k\}, \\ 0 & \text{otherwise.} \end{cases}$$

   where $w = \exp(-2\mathbf{i}\pi/N)$. Thus $x = \hat{x}$.

4.b One has $|\mathrm{supp}(x)| = k$, hence $|\mathrm{supp}(x)| \times |\mathrm{supp}(\hat{x})| = k^2 = N$.

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Exercise 3.3 (Sparsity of Basis Pursuit)
Let $A \in \mathbb{R}^{m \times N}$ **and** $y \in \mathbb{R}^m$**. Assume that the problem**

$$\mathbf{argmin}_{At=y} \|t\|_1 = \{\hat{x}\} \tag{8}$$

has a unique solution. Denote by $a_1, \cdots, a_N$ **the columns of** A **in** $\mathbb{R}^m$**.**

1. **Show that the family $\{a_j : j \in \mathrm{supp}(\hat{x})\}$ is linearly independent.**

2. **Deduce that $\|\hat{x}\|_0 \leq m$.**

****

Solution of the exercise 3.3

1. Argue by contradiction. Let $J_{\hat{x}} \subset \{1, \dots, N\}$ be the support of $\hat{x}$, and assume that $\{a_j : j \in J_{\hat{x}}\}$ is linearly dependent. Let $v \in \mathbb{R}^{J_{\hat{x}}}$ be nonzero and such that $\sum_{j \in J_{\hat{x}}} v_j a_j = 0$.

   By uniqueness, for every $t \neq 0$ one has $\|\hat{x}\|_1 < \|\hat{x} + tv\|_1$, since $Av = 0$ and therefore $A(\hat{x}+tv) = A\hat{x} = y$.

   On the other hand,
   $$\|\hat{x} + tv\|_1 = \sum_{j \in J_{\hat{x}}} \mathrm{sgn}(\hat{x}_j + tv_j)(\hat{x}_j + tv_j).$$
   For $|t| < \min_{j \in J_{\hat{x}}} |\hat{x}_j| / \|v\|_\infty$, one has $\mathrm{sgn}(\hat{x}_j + tv_j) = \mathrm{sgn}(\hat{x}_j)$, hence
   $$\|\hat{x} + tv\|_1 = \|\hat{x}\|_1 + t \sum_j v_j \mathrm{sgn}(\hat{x}_j).$$
   Since $\|\hat{x} + tv\|_1 > \|\hat{x}\|_1$, one gets $t \sum_j v_j \mathrm{sgn}(\hat{x}_j) > 0$ for every such t. This is impossible.

2. $\{a_j : j \in \mathrm{supp}(\hat{x})\}$ is a linearly independent family of vectors in $\mathbb{R}^m$, so it has at most m elements. Thus $\|\hat{x}\|_0 \leq m$.

****

Exercise 3.4 (RIP and the minimal number of measurements)
Let $A \in \mathbb{R}^{m \times N}$ be such that for every $x \in \Sigma_{2s}$, $(1/2)\|x\|_2 \leq \|Ax\|_2 \leq (3/2)\|x\|_2$. Then there exist absolute constants $c_0, c_1 > 0$ such that $m \geq c_0 s \log(c_1 N/s)$.

****

Solution of the exercise 3.4 For every $x \in \Sigma_{2s} \cap B_2^N$, one has $Ax \in (3/2)B_2^m$. Moreover, for all $x, y \in \Sigma_s \cap B_2^N$, if $\|x - y\|_2 \geq \varepsilon$ then $\|Ax - Ay\|_2 \geq (1/2)\|x - y\|_2 \geq \varepsilon/2$.

Consequently, if Λ_ε is an ε-separated subset of $\Sigma_s \cap B_2^N$ for ℓ_2^N, then $A\Lambda_\varepsilon$ is an $(\varepsilon/2)$-separated subset of $(3/2)B_2^m$ for ℓ_2^m.

One can construct a $(1/4)$-separated subset of $\Sigma_s \cap B_2^N$ for ℓ_2^N with cardinality at least $(c_0 N/s)^{s/c_1}$, where c_0, c_1 are absolute constants. Thus $(3/2)B_2^m$ contains a $1/8$-separated set for ℓ_2^m of cardinality $(c_0 N/s)^{s/c_1}$. Any such set has cardinality at most 25^m. Indeed, after rescaling by $2/3$, the set is $(1/12)$-separated in B_2^m, and the volumetric argument bounds its cardinality by $(1 + 2 \cdot 12)^m = 25^m$.

Note: $RIP(65s)$ implies $RE(s)$, and $RE(s)$ implies $m \geq c_0 s \log(c_1 N/s)$.

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Exercise 3.5 (Robustness and the lower RIP bound)
Let $A \in \mathbb{R}^{m \times N}$ be a measurement matrix and $\Delta : \mathbb{R}^m \to \mathbb{R}^N$ a decompression map (not

necessarily linear). One says that (A, Δ) is C-robust of order s when, for every $x \in \Sigma_s$ and every $e \in \mathbb{R}^m$,

$$\|\Delta(Ax + e) - x\|_2 \leq C \|e\|_2 .$$

In particular, if (A, Δ) is C-robust of order s, then $\Delta(Ax) = x$ for every $x \in \Sigma_s$. Thus Δ reconstructs exactly every $x \in \Sigma_s$ from Ax. Robustness asks in addition that Δ tolerate errors in the observations: the vector $\Delta(Ax+e)$ reconstructed from the noisy observations $Ax + e$ stays close to x, up to a term of the order of the noise.

Show that if (A, Δ) is C-robust of order s, then the lower RIP bound of order $2s$ is necessary: for every $x \in \Sigma_{2s}$,

$$\|Ax\|_2 \geq \frac{1}{C} \|x\|_2 .$$

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Solution of the exercise 3.5 Let $x, y \in \Sigma_s$. Set

$$e_y = \frac{A(x-y)}{2} \text{ and } e_x = \frac{A(y-x)}{2}.$$

Then $Ax + e_x = Ay + e_y$, so $\hat{x} = \Delta(Ax + e_x) = \Delta(Ay + e_y) = \hat{y}$. Therefore

$$\|x - y\|_2 = \|x - \hat{x} + \hat{y} - y\|_2 \leq \|x - \hat{x}\|_2 + \|y - \hat{y}\|_2 \leq C \|e_x\|_2 + C \|e_y\|_2 = C \|A(x - y)\|_2 .$$

****

Exercise 3.6 (ℓ_2-stability)
Let $A \in \mathbb{R}^{m \times N}$ be a measurement matrix and $\Delta : \mathbb{R}^m \to \mathbb{R}^N$ a decompression procedure. One says that (A, Δ) is C-ℓ_2-stable of order s when, for every $x \in \mathbb{R}^N$,

$$\|x - \Delta(Ax)\|_2 \leq C\sigma_{s,2}(x) \text{ where } \sigma_{s,2}(x) = \min_{z \in \Sigma_s} \|x - z\|_2 . \tag{9}$$

We will show that if there exists a pair (A, Δ) which is C-ℓ_2-stable of order s (for any s, even $s = 1$), then necessarily $m \geq \big(1 - \sqrt{C^2 - 1}/C\big)N$, where one assumes $C \geq 1$.

1. **Show that for every $x \in \ker(A)$ and every $j \in \{1, \dots, N\}$, $|x_j| \leq C' \|x\|_2$ with $C' = \sqrt{C^2 - 1}/C$.**

2. **Let L be a linear subspace of $\mathbb{R}^N$ such that for every $x \in L$ and every $i \in \{1, \dots, N\}$, $|x_i| \leq C' \|x\|_2$. Show that $\dim(L) \leq C'N$.**

3. **Deduce that $m \geq (1 - \sqrt{C^2 - 1}/C)N$.**

Note: Stability matters when the signals to be reconstructed are not exactly sparse, but only close to Σ_s. Such signals are called *compressible*. This exercise shows that (9) cannot hold unless the number of measurements is of order N, which is not the compressed-sensing regime. This is why stability with respect to $\sigma_{s,1}(x)$ was introduced.

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Solution of the exercise 3.6

1. One has $\Delta(0) = 0$, so for every $x \in \ker(A)$, $\|x\|_2 \leq C\sigma_{s,2}(x)$. For $s = 1$, for every $x \in \ker(A)$ and every $j = 1, \dots, N$,
$$\|x\|_2^2 \leq C^2\big(\|x\|_2^2 - |x_j|^2\big)$$
hence $|x_j|^2 \leq (C^2-1)/C^2 \|x\|_2^2$, that is $|x_j| \leq C' \|x\|_2$ with $C' = \sqrt{C^2-1}/C$.

2. Let P be the orthogonal projection onto L, and let $(e_1, \dots, e_N)$ be the canonical basis of $\mathbb{R}^N$. For every j, $Pe_j \in L$, so $|(Pe_j)_j| \leq C' \|Pe_j\|_2$. Then
$$\dim(L) = \mathrm{Tr}(P) = \sum_{j=1}^N \langle Pe_j, e_j\rangle \leq \sum_{j=1}^N C' \|Pe_j\|_2 \leq C'N.$$

3. Therefore $N - m \geq \dim(\ker(A)) \leq C'N$, and $m \geq \big(1 - \sqrt{C^2-1}/C\big)N$.

****

Exercise 3.7 (Incoherence and the Welch bound)
Let $A \in \mathbb{R}^{m\times N}$. The *coherence of* A is
$$\mu(A) = \max_{1\leq i<j\leq N} \frac{|\langle a_i, a_j\rangle|}{\|a_i\|_2 \|a_j\|_2},$$
where $a_1, \dots, a_N$ are the columns of A.

1. **Assume that the columns of A have ℓ_2^m-norm equal to 1. Show that A satisfies RIP(s) whenever $\mu(A) < 1/s$.**

2. **Welch bound: show that if $N \geq 2m-1$, then for every matrix A, $\mu(A) \geq 1/\sqrt{2m}$.**

In particular, the Welch bound shows that one cannot prove RIP from the incoherence of a matrix when the number of measurements m is smaller than $s^2/2$. This approach (showing that the coherence of A satisfies $\mu(A) < 1/s$) therefore cannot reach the minimal number of measurements of order $s\log(ep/s)$. On the other hand, one can construct deterministic matrices with an optimal number of rows, of order s^2 as given by the Welch bound, whose incoherence is of order $1/s$.

Geometrically, building a matrix that meets the Welch bound for the smallest possible coherence amounts to finding N vectors in dimension m on the unit sphere of ℓ_2^m whose mutual angles are as large as possible. Such sets tend to have essentially the same angle between every pair. They are called *equiangular*. A deterministic construction of an equiangular set of maximal cardinality yields deterministic RIP matrices. These matrices necessarily have a number of rows of order s^2, which is much larger than the $s\log(ep/s)$ obtained by random constructions.

****

Solution of the exercise 3.7

1. Let $x \in \Sigma_s$. Since $\langle a_i, a_i \rangle = 1$,

$$\big|\, \|Ax\|_2^2 - \|x\|_2^2 \,\big| = \Big| \sum_{i \neq j} x_i x_j \langle a_i, a_j \rangle \Big| \leq \mu(A) \sum_{i \neq j} |x_i x_j| \leq \mu(A) \Big(\sum_i |x_i| \Big)^2 \leq s\mu(A) \, \|x\|_2^2 \,.$$

Hence, for every $x \in \Sigma_s$,

$$(1 - s\mu(A)) \, \|x\|_2^2 \leq \|Ax\|_2^2 \leq (1 + s\mu(A)) \, \|x\|_2^2 \,.$$

This is a RIP condition of order s as soon as $\mu(A) < 1/s$.

2. One may assume that every vector a_i has norm 1. Consider the Gram matrix of $a_1, \ldots, a_N$, namely $G = \big(\langle a_i, a_j \rangle\big)_{1 \leq i,j \leq N}$. It is symmetric and positive semidefinite, so its trace equals its nuclear norm (the sum of its singular values):

$$\|G\|_{S_1} = \mathrm{Tr}(G) = \sum_{j=1}^{N} \langle a_j, a_j \rangle = N.$$

The rank of G is at most m, since

$$G = XX^\top \text{ where } X = \begin{bmatrix} a_1^\top \\ \vdots \\ a_N^\top \end{bmatrix}.$$

Thus $\|G\|_{S_1} \leq \sqrt{m} \, \|G\|_{S_2}$, where

$$\|G\|_{S_2}^2 = \sum_{i,j} \langle a_i, a_j \rangle^2.$$

Therefore

$$\sum_{i,j} \langle a_i, a_j \rangle^2 = N + \sum_{i \neq j} \langle a_i, a_j \rangle^2 \geq \frac{N^2}{m},$$

and

$$\sum_{i \neq j} \langle a_i, a_j \rangle^2 \geq \frac{N(N-m)}{m}.$$

Moreover,

$$\mu^2(A) \geq \frac{1}{N(N-1)} \sum_{i \neq j} \langle a_i, a_j \rangle^2 \geq \frac{N-m}{m(N-1)}.$$

When $N \geq 2m - 1$, the right-hand side is at least $1/(2m)$, which is the claim.

4 Optimization

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Exercise 4.1 (Convex optimization under linear constraints)

Consider the optimization problem

$$\min \Big(x^2 + yz + y^2 + z^2 : x \leq -1, z \leq -1 \Big).$$

1. **Reduce this minimization problem to a problem of the form**

$$\min_{t\in K} f(t) \qquad (P)$$

where f is real-valued and $K = \{t : g(t) \leq 0\}$ with $g : \mathbb{R}^2 \to \mathbb{R}$ convex and of class $\mathcal{C}^1$.

2. **Show that f is convex.**

3. **Show that the constraint K is qualified.**

4. **Write the Lagrangian of (P).**

5. **Write the KKT conditions of (P) and solve (P) from these equations.**

6. **Apply the duality theorem. Write the dual problem of (P), and recover the solution of (P) obtained in question 5).**

****

Solution of the exercise 4.1

1. The objective is easily minimized in x, and the minimum is attained at $x = -1$. It remains to minimize in y and z. One is led to the problem

$$(P) \qquad \min_{t\in K} f(t)$$

where $f : \mathbb{R}^2 \to \mathbb{R}$ is given by $f(y, z) = yz + y^2 + z^2$ and $K = \{(y, z)^\top \in \mathbb{R}^2 : g(y, z) \leq 0\}$ with $g(y, z) = z + 1$, which is convex and of class $\mathbb{C}^1$.

2. One has $f(y, z) = (y + z/2)^2 + 3z^2/4$. This is a sum of squares of linear forms, hence a convex function.

3. Let $(y, z)^\top \in K$. Then $\nabla g(y, z) = (0, 1)^\top$, so for $v = (0, -1)^\top$ one has $\langle \nabla g(y, z), v\rangle < 0$. Thus K is qualified at $(y, z)^\top \in K$. Since this holds at every point of K, the constraint K is qualified.

4. The Lagrangian of (P) is, for every $(y, z)^\top \in \mathbb{R}^2$ and $\lambda \geq 0$,

$$L((y, z), \lambda) = f(y, z) + \lambda g(y, z) = yz + y^2 + z^2 + \lambda(z + 1).$$

5. The KKT conditions are: if $(y^*, z^*)^\top \in \mathbb{R}^2$ solves (P), then there exists $\lambda^* \geq 0$ such that

   (a) $(y^*, z^*)^\top \in K$,

   (b) $\lambda^* g(y^*, z^*) = 0$,

   (c) $\nabla_{(y,z)} L((y^*, z^*), \lambda^*) = 0$.

   One obtains $y^* = 1/2$ and $z^* = -1$. The value of the original optimization problem is therefore 1.75, attained at $(-1, 1/2, -1)^\top$.

6. The duality theorem says that

$$\min_{t\in K} f(t) = \min_{t\in\mathbb{R}^2}\sup_{\lambda\geq 0} L(t,\lambda) = \sup_{\lambda\geq 0}\min_{t\in\mathbb{R}^2} L(t,\lambda).$$

Moreover, if $\lambda^* \geq 0$ solves the dual problem

$$\sup_{\lambda\geq 0} d(\lambda) \text{ where } d(\lambda) = \min_{t\in\mathbb{R}^2} L(t,\lambda) = \frac{-\lambda^2}{3} + \lambda,$$

then a solution t^* of (P) also solves

$$\min_{t\in\mathbb{R}^2} L(t,\lambda^*).$$

One finds that $\lambda^* = 3/2$ solves the dual problem, and then $t^* = (1/2,-1)^\top$ solves the primal problem.

****

Exercise 4.2 (Linear systems and optimization)
Let A be a symmetric positive definite matrix in $\mathbb{R}^{N\times N}$ and let $y\in\mathbb{R}^N$. Show that the following two assertions are equivalent:

1. **$\hat{x}$ solves $Ax = y$;**
2. **$\hat{x}$ minimizes $x \mapsto F(x) = \frac{1}{2}x^\top Ax - x^\top y$.**

****

Solution of the exercise 4.2

First solution: The Hessian of F is A, which is symmetric positive definite. Thus $\hat{x}$ minimizes F if and only if $\nabla F(\hat{x}) = 0$. And $\nabla F(x) = Ax - y$.

Second solution: If $Ax_0 = y$, then for every $x\in\mathbb{R}^N$,

$$F(x) - F(x_0) = \tfrac{1}{2}x^\top Ax - x^\top y - \tfrac{1}{2}x_0^\top Ax_0 + x_0^\top y = \tfrac{1}{2}\Big(x^\top Ax + x_0^\top Ax_0 - 2x^\top Ax_0\Big). \tag{10}$$

By Cauchy–Schwarz,

$$2|x^\top Ax_0| = 2|\langle A^{1/2}x, A^{1/2}x_0\rangle| \leq 2\left\|A^{1/2}x\right\|_2\left\|A^{1/2}x_0\right\|_2 \leq \left\|A^{1/2}x\right\|_2^2 + \left\|A^{1/2}x_0\right\|_2^2 = x^\top Ax + x_0^\top Ax_0.$$

Hence $F(x) - F(x_0) \geq 0$, so x_0 minimizes F.

Conversely, if $F(x) - F(x_0) \geq 0$ for every x, then for every $h\in\mathbb{R}^N$,

$$F(x_0+h) = F(x_0) + \big\langle \nabla F(x_0), h\big\rangle + \frac{1}{2}h^\top\nabla^2 F(x_0)h = F(x_0) + \big\langle Ax_0 - y, h\big\rangle + \frac{\left\|A^{1/2}h\right\|_2^2}{2}$$

and since $F(x_0+h)\geq F(x_0)$, one has $2\big\langle Ax_0 - y, h\big\rangle + \left\|A^{1/2}h\right\|_2^2 \geq 0$ for every h. Replacing h by λh for $\lambda > 0$ shows that $\big\langle Ax_0 - y, h\big\rangle \geq 0$ for every h, hence $Ax_0 = y$.

****

Exercise 4.3 (Gradient descent)
Consider the function

$$\varphi(t) = \frac{t}{\sqrt{1+t^2}}.$$

Note that $\varphi(x^*) = 0$ if and only if $x^* = 0$. Describe the convergence of Newton's method for the equation $\varphi(x) = 0$.

****

Solution of the exercise 4.3 For every $t \in \mathbb{R}$,

$$\varphi'(t) = (1+t^2)^{-3/2}.$$

Newton's method is the sequence $(t_k)_k$ starting from some $t_0 \in \mathbb{R}$, given by

$$t_{k+1} = t_k - \frac{\varphi(t_k)}{\varphi'(t_k)} = -t_k^3.$$

Therefore

1. $(t_k)_k$ converges to 0 when $|t_0| < 1$;
2. $(t_k)_k$ oscillates between -1 and 1 when $|t_0| = 1$;
3. $(t_k)_k$ diverges when $|t_0| > 1$.

****

Exercise 4.4 (Characterization of strong convexity)
Let $f : (a,b) \to \mathbb{R}$. One says that f is strongly convex with convexity modulus $c > 0$ when, for all $x, y \in (a,b)$ and every $t \in [0,1]$,

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) - ct(1-t)(x-y)^2.$$

Show that

1. **f is strongly convex with modulus $c > 0$ if and only if, for every $x_0 \in (a,b)$ and every x,**
$$f(x) \geq f(x_0) + L(x - x_0) + c(x - x_0)^2$$
where $L \in [f'_-(x_0), f'_+(x_0)]$ lies between the left and right derivatives of f at x_0.

2. **If f is differentiable, then f is strongly convex with modulus $c > 0$ if and only if, for all $x, y \in (a,b)$,**
$$(f'(x) - f'(y))(x - y) \geq 2c(x-y)^2.$$

3. **If f is twice differentiable, then f is strongly convex with modulus $c > 0$ if and only if, for every $x \in (a,b)$,**
$$f''(x) \geq 2c.$$

****

Solution of the exercise 4.4 See "Some Characterizations of Strongly Convex Functions in Inner Product Spaces" by Teodoro Lara, Merentes, Rosales and Valear, in *Mathematica Aeterna*.

****

Exercise 4.5 (Subdifferential of a differentiable function)

Show that if $f : \mathbb{R}^n \to \mathbb{R}$ is a convex function that is differentiable at x, then $\partial^- f(x) = \{\nabla f(x)\}$.

****

Solution of the exercise 4.5 Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function, differentiable at x. We want to show that the subdifferential of f at x, denoted by $\partial^- f(x)$, consists exactly of the singleton containing the gradient $\nabla f(x)$. We will prove this by showing mutual inclusion.

Part 1: $\partial^- f(x) \subseteq \{\nabla f(x)\}$

Let $g \in \partial^- f(x)$. By the definition of the subgradient, for any $y \in \mathbb{R}^n$, we have:

$$f(y) \geq f(x) + \langle g, y - x \rangle$$

Let $v \in \mathbb{R}^n$ be an arbitrary direction, and let $t > 0$ be a scalar. Substituting $y = x + tv$ into the subgradient inequality yields:

$$f(x + tv) \geq f(x) + t\langle g, v \rangle$$

Rearranging this inequality, we obtain:

$$\frac{f(x + tv) - f(x)}{t} \geq \langle g, v \rangle$$

Since f is differentiable at x, the directional derivative exists and can be expressed via the gradient as $\langle \nabla f(x), v \rangle$. Taking the limit as $t \to 0^+$ gives:

$$\langle \nabla f(x), v \rangle \geq \langle g, v \rangle \implies \langle \nabla f(x) - g, v \rangle \geq 0$$

Because this inequality holds for all $v \in \mathbb{R}^n$, we are free to choose $v = g - \nabla f(x)$. This specific choice yields:

$$\langle \nabla f(x) - g, g - \nabla f(x) \rangle \geq 0 \implies -\|\nabla f(x) - g\|^2 \geq 0$$

Since the squared norm of any real vector must be non-negative, we must have $\|\nabla f(x) - g\|^2 = 0$, which immediately implies $g = \nabla f(x)$. Therefore, the only possible subgradient is the gradient, meaning $\partial^- f(x) \subseteq \{\nabla f(x)\}$.

Part 2: $\{\nabla f(x)\} \subseteq \partial^- f(x)$

Since f is a convex function and is differentiable at x, the first-order characterization of convexity dictates that the tangent hyperplane lies entirely below or on the graph of the function. That is, for all $y \in \mathbb{R}^n$:

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$$

This inequality matches the exact definition of a subgradient at x. Thus, the gradient itself is a valid subgradient, meaning $\nabla f(x) \in \partial^- f(x)$.

Conclusion

Combining both inclusions, we definitively conclude that:

$$\partial^- f(x) = \{\nabla f(x)\}$$

****

Exercise 4.6 (Subdifferential of the absolute value)

Compute the subdifferential of the absolute value function $f(x) = |x|$ over $\mathbb{R}$.

****

Solution of the exercise 4.6 The function $f(x) = |x|$ is convex on $\mathbb{R}$. To compute its subdifferential $\partial f(x)$, we separate the problem into three cases based on the value of x:

Case 1: $x > 0$

For $x > 0$, the function simplifies to $f(x) = x$. In this open domain, f is strictly differentiable with derivative $f'(x) = 1$. Since the subdifferential of a convex, differentiable function is just the singleton of its derivative, we have:

$$\partial f(x) = \{1\}$$

Case 2: $x < 0$

For $x < 0$, the function simplifies to $f(x) = -x$. Again, f is differentiable here with derivative $f'(x) = -1$. Thus:

$$\partial f(x) = \{-1\}$$

Case 3: $x = 0$

At $x = 0$, the function is not differentiable, so we must use the fundamental definition of the subdifferential. A scalar $g \in \partial f(0)$ if and only if for all $y \in \mathbb{R}$:

$$f(y) \geq f(0) + g(y - 0)$$

Substituting $f(y) = |y|$ and $f(0) = 0$, the condition becomes:

$$|y| \geq gy$$

This inequality must hold for all possible values of $y \in \mathbb{R}$.

- If we choose $y > 0$, dividing by y yields $1 \geq g$.
- If we choose $y < 0$, dividing by y (which reverses the inequality) yields $-1 \leq g$.

Combining these two requirements, we find that any g in the closed interval $[-1, 1]$ satisfies the definition. Thus:

$$\partial f(0) = [-1, 1]$$

Conclusion

Summarizing all three cases, the subdifferential of the absolute value function $f(x) = |x|$ is given by:

$$\partial f(x) = \begin{cases} \{-1\} & \text{if } x < 0 \\ [-1, 1] & \text{if } x = 0 \\ \{1\} & \text{if } x > 0 \end{cases}$$

****

Exercise 4.7 (Subdifferential and minimization)

Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function. Recall that the subdifferential of f at x is

$$\partial^- f(x) = \big\{ g \in \mathbb{R} : f(y) \geq f(x) + \big\langle g, y - x \big\rangle \text{ for every } y \in \mathbb{R} \big\}.$$

Show that x^* solves $\min_x f(x)$ if and only if $0 \in \partial^- f(x^*)$.

****

Solution of the exercise 4.7 Assume that $f(x) \geq f(x^*)$ for every $x \in \mathbb{R}$. Then for every y, $f(y) \geq f(x^*) + \big\langle 0, y - x \big\rangle$, so $0 \in \partial^- f(x^*)$.

Conversely, if $0 \in \partial^- f(x^*)$, then for every y, $f(y) \geq f(x^*) + \big\langle 0, y - x \big\rangle = f(x^*)$. Thus x^* solves the minimization problem $\min_x f(x)$.

****

Exercise 4.8 (Subdifferential of the Euclidean norm)

Let $f : \mathbb{R}^n \to \mathbb{R}$ be the function defined by $f(x) = \|x\|_2$, where $\| \cdot \|_2$ denotes the standard Euclidean norm. Determine the subdifferential $\partial f(x)$ for all $x \in \mathbb{R}^n$.

****

Solution of the exercise 4.8 We consider two cases: $x \neq 0$ and $x = 0$.

Case 1: $x \neq 0$

The function $f(x) = \|x\|_2$ is differentiable at any non-zero $x \in \mathbb{R}^n$. Since f is convex and differentiable at $x \neq 0$, its subdifferential is a singleton containing only its gradient. Using the chain rule, we compute the gradient:

$$\nabla f(x) = \frac{x}{\|x\|_2}$$

Thus, for $x \neq 0$, we have:

$$\partial f(x) = \left\{ \frac{x}{\|x\|_2} \right\}$$

Case 2: $x = 0$

By definition, a vector $g \in \mathbb{R}^n$ belongs to the subdifferential $\partial f(0)$ if and only if for all $y \in \mathbb{R}^n$:

$$f(y) \geq f(0) + \langle g, y - 0 \rangle$$

Substituting $f(y) = \|y\|_2$ and $f(0) = 0$, this condition becomes:

$$\|y\|_2 \geq \langle g, y \rangle \quad \text{for all } y \in \mathbb{R}^n.$$

We now show that this is equivalent to $\|g\|_2 \leq 1$.

First, assume $\|g\|_2 \leq 1$*:*

By the Cauchy-Schwarz inequality, for any $y \in \mathbb{R}^n$:

$$\langle g, y \rangle \leq \|g\|_2 \|y\|_2 \leq 1 \cdot \|y\|_2 = \|y\|_2$$

This implies that any g in the closed unit ball belongs to $\partial f(0)$.

Next, assume $g \in \partial f(0)$*:*

The inequality $\|y\|_2 \geq \langle g, y \rangle$ holds for all $y \in \mathbb{R}^n$. If we choose $y = g$, we get:

$$\|g\|_2 \geq \langle g, g \rangle = \|g\|_2^2$$

Dividing by $\|g\|_2 > 0$ (assuming $g \neq 0$), we obtain $1 \geq \|g\|_2$. (If $g = 0$, the inequality $\|g\|_2 \leq 1$ trivially holds).

Thus, for $x = 0$, the subdifferential is the closed Euclidean unit ball:

$$\partial f(0) = \{g \in \mathbb{R}^n \mid \|g\|_2 \leq 1\}$$

****

Exercise 4.9 (Proximal operator of an indicator function (Projection))

Let $C \subset \mathbb{R}^n$ be a non-empty closed convex set, and let ι_C denote its indicator function:

$$\iota_C(u) = \begin{cases} 0 & \textbf{if } u \in C, \\ +\infty & \textbf{if } u \notin C. \end{cases}$$

Show that for any parameter $\gamma > 0$ and any $x \in \mathbb{R}^n$, the proximal operator of $\gamma\iota_C$ is the orthogonal projection onto C, i.e.,

$$\text{prox}_{\gamma\iota_C}(x) = \Pi_C(x).$$

****

Solution of the exercise 4.9 By definition of the proximal operator:

$$\text{prox}_{\gamma\iota_C}(x) = \arg\min_{u \in \mathbb{R}^n} \left\{ \gamma\iota_C(u) + \frac{1}{2}\|u - x\|_2^2 \right\}.$$

Since $\gamma > 0$, we have $\gamma\iota_C(u) = \iota_C(u)$. Therefore, the objective function evaluates to:

$$\iota_C(u) + \frac{1}{2}\|u - x\|_2^2 = \begin{cases} \frac{1}{2}\|u - x\|_2^2 & \text{if } u \in C, \\ +\infty & \text{if } u \notin C. \end{cases}$$

Minimizing this expression over $\mathbb{R}^n$ is equivalent to minimizing the squared Euclidean distance over C:

$$\text{prox}_{\gamma\iota_C}(x) = \arg\min_{u\in C} \frac{1}{2}\|u - x\|_2^2.$$

By definition, the unique point in C that minimizes the Euclidean distance to x is the projection $\Pi_C(x)$. Thus:

$$\text{prox}_{\gamma\iota_C}(x) = \Pi_C(x).$$

Notice that this operator is independent of the step size $\gamma > 0$.

****

Exercise 4.10 (Proximal operator of a convex quadratic function)

Let $A \in \mathbb{R}^{n\times n}$ be a symmetric positive semi-definite matrix, $b \in \mathbb{R}^n$, and let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by:

$$f(u) = \frac{1}{2}\langle u, Au\rangle - \langle b, u\rangle.$$

Show that for any $\gamma > 0$ and any $x \in \mathbb{R}^n$, the proximal operator $\text{prox}_{\gamma f}(x)$ is given by:

$$\text{prox}_{\gamma f}(x) = (I_n + \gamma A)^{-1}(x + \gamma b),$$

where I_n is the $n \times n$ identity matrix.

****

Solution of the exercise 4.10 Since A is symmetric positive semi-definite, f is convex and differentiable with gradient:

$$\nabla f(u) = Au - b.$$

For $\gamma > 0$, the proximal objective:

$$\phi(u) = \frac{1}{2}\langle u, Au\rangle - \langle b, u\rangle + \frac{1}{2\gamma}\|u - x\|_2^2$$

is strongly convex with parameter at least $1/\gamma > 0$. The unique minimizer $p = \text{prox}_{\gamma f}(x)$ satisfies $\nabla\phi(p) = 0$:

$$Au - b + \frac{1}{\gamma}(p - x) = 0.$$

Multiplying through by γ yields:

$$\gamma(Ap - b) + p - x = 0 \iff (I_n + \gamma A)p = x + \gamma b.$$

Because A is symmetric positive semi-definite, its eigenvalues satisfy $\lambda_i(A) \geq 0$. Thus, the eigenvalues of $I_n + \gamma A$ satisfy:

$$\lambda_i(I_n + \gamma A) = 1 + \gamma\lambda_i(A) \geq 1 > 0.$$

The matrix $(I_n + \gamma A)$ is strictly positive definite and invertible. Multiplying by $(I_n + \gamma A)^{-1}$ gives:

$$\text{prox}_{\gamma f}(x) = (I_n + \gamma A)^{-1}(x + \gamma b).$$

****

Exercise 4.11 (Soft-thresholding operator (ℓ_1-norm proximal operator))
Let $f : \mathbb{R} \to \mathbb{R}$ be the absolute value function $f(u) = |u|$, and let $\gamma > 0$.

1. **State the subdifferential inclusion characterizing $p = \text{prox}_{\gamma f}(x)$.**
2. **Solve for p across all cases and conclude that:**

$$\text{prox}_{\gamma f}(x) = \text{sign}(x) \max(|x| - \gamma, 0).$$

****

Solution of the exercise 4.11 1. Optimality condition:
By definition:

$$p = \text{prox}_{\gamma f}(x) = \arg\min_{u \in \mathbb{R}} \left\{ |u| + \frac{1}{2\gamma}(u - x)^2 \right\}.$$

Since the objective is strictly convex, the minimizer p is uniquely characterized by:

$$0 \in \partial|p| + \frac{1}{\gamma}(p - x) \iff x - p \in \gamma \partial|p|.$$

Recall that the subdifferential of the absolute value is:

$$\partial|p| = \begin{cases} \{1\} & \text{if } p > 0, \\ \{-1\} & \text{if } p < 0, \\ [-1, 1] & \text{if } p = 0. \end{cases}$$

2. Derivation:
We evaluate the three possible cases:

- If $p > 0$: $x - p = \gamma \implies p = x - \gamma$. This holds if and only if $x > \gamma$.
- If $p < 0$: $x - p = -\gamma \implies p = x + \gamma$. This holds if and only if $x < -\gamma$.
- If $p = 0$: $x - 0 \in [-\gamma, \gamma] \implies |x| \leq \gamma$.

Combining the cases:

$$\text{prox}_{\gamma f}(x) = \begin{cases} x - \gamma & \text{if } x > \gamma, \\ 0 & \text{if } |x| \leq \gamma, \\ x + \gamma & \text{if } x < -\gamma, \end{cases}$$

which is compactly written as:

$$\text{prox}_{\gamma f}(x) = \text{sign}(x) \max(|x| - \gamma, 0).$$

****

Exercise 4.12 (Hard-thresholding operator (ℓ_0 pseudo-norm))

Let $\lambda > 0$ and let $f : \mathbb{R} \to \mathbb{R}$ be the non-convex counting penalty defined by:

$$f(u) = \lambda \|u\|_0 = \begin{cases} 0 & \text{if } u = 0, \\ \lambda & \text{if } u \neq 0. \end{cases}$$

For a parameter $\gamma > 0$, the proximal operator of γf at $x \in \mathbb{R}$ is defined as the set of minimizers:

$$\text{prox}_{\gamma f}(x) = \arg\min_{u \in \mathbb{R}} \left\{ \lambda \|u\|_0 + \frac{1}{2\gamma}(u - x)^2 \right\}.$$

1. **Evaluate the infimum of the proximal objective function separately on $\{0\}$ and on $\mathbb{R} \setminus \{0\}$.**
2. **Deduce the explicit expression of $\text{prox}_{\gamma f}(x)$ for every $x \in \mathbb{R}$.**
3. **Compare this operator with convex proximal operators regarding uniqueness and continuity.**

****

Solution of the exercise 4.12 Let $\phi(u) = \lambda \|u\|_0 + \frac{1}{2\gamma}(u - x)^2$ denote the objective function to minimize over $\mathbb{R}$.

1. Candidate minimizers on the partitioned domain:

We decompose $\mathbb{R} = \{0\} \cup (\mathbb{R} \setminus \{0\})$:

- **At $u = 0$:** Since $\|0\|_0 = 0$, the objective value is:

$$\phi(0) = \frac{1}{2\gamma} x^2.$$

- **On $u \neq 0$:** For all $u \in \mathbb{R} \setminus \{0\}$, $\|u\|_0 = 1$, which gives:

$$\phi(u) = \lambda + \frac{1}{2\gamma}(u - x)^2.$$

We minimize ϕ on $\mathbb{R} \setminus \{0\}$:

  - If $x \neq 0$, the unconstrained minimizer $u = x$ belongs to $\mathbb{R} \setminus \{0\}$, yielding:

$$\inf_{u \neq 0} \phi(u) = \phi(x) = \lambda.$$

  - If $x = 0$, $\phi(u) = \lambda + \frac{1}{2\gamma} u^2 > \lambda$ for all $u \neq 0$, so the infimum λ is not attained on $\mathbb{R} \setminus \{0\}$.

2. Global minimization and derivation of $\text{prox}_{\gamma f}(x)$:

Any global minimizer is either $u = 0$ or the point achieving the minimum on $\mathbb{R} \setminus \{0\}$, which is $u = x$. Comparing the two values:

$$\phi(0) - \phi(x) = \frac{1}{2\gamma} x^2 - \lambda.$$

We distinguish three cases based on the sign of this difference:

- **Case $|x| < \sqrt{2\gamma\lambda}$:**
  $\frac{1}{2\gamma}x^2 < \lambda \implies \phi(0) < \phi(x)$. The unique minimizer is $u = 0$.
- **Case $|x| > \sqrt{2\gamma\lambda}$:**
  $\frac{1}{2\gamma}x^2 > \lambda \implies \phi(x) < \phi(0)$. The unique minimizer is $u = x$.
- **Case $|x| = \sqrt{2\gamma\lambda}$:**
  $\phi(0) = \phi(x) = \lambda$. Both points achieve the global minimum simultaneously.

Therefore, the proximal mapping is given by:

$$\text{prox}_{\gamma f}(x) = \begin{cases} \{0\} & \text{if } |x| < \sqrt{2\gamma\lambda}, \\ \{0, x\} & \text{if } |x| = \sqrt{2\gamma\lambda}, \\ \{x\} & \text{if } |x| > \sqrt{2\gamma\lambda}. \end{cases}$$

3. Differences from convex proximal operators:

- **Multi-valuedness:** When f is convex, the objective $u \mapsto f(u) + \frac{1}{2\gamma}\|u - x\|_2^2$ is strongly convex, guaranteeing a unique global minimizer for every x. Here, at the boundary points $|x| = \sqrt{2\gamma\lambda}$, the minimizer is non-unique and $\text{prox}_{\gamma f}(x)$ is set-valued.
- **Discontinuity (Thresholding Jump):** Selecting a single-valued branch by convention (e.g., choosing 0 at $|x| = \sqrt{2\gamma\lambda}$) yields the classical hard-thresholding operator:

  $$H_{\sqrt{2\gamma\lambda}}(x) = \begin{cases} 0 & \text{if } |x| \leq \sqrt{2\gamma\lambda}, \\ x & \text{if } |x| > \sqrt{2\gamma\lambda}. \end{cases}$$

  This operator is discontinuous at $x = \pm\sqrt{2\gamma\lambda}$ (jumping by $\sqrt{2\gamma\lambda}$), whereas convex proximal operators are always firmly non-expansive, continuous, and 1-Lipschitz.

****

Exercise 4.13 (Proximal Gradient Descent and the ISTA Algorithm)

Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex and continuously differentiable, and let $g : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be a proper, lower semicontinuous convex function. Consider the composite optimization problem:

$$\min_{x \in \mathbb{R}^n} F(x) := f(x) + g(x).$$

For any step size $\gamma > 0$, the proximal gradient algorithm (Forward-Backward splitting) is defined by:

$$x_{k+1} = \text{prox}_{\gamma g}\left(x_k - \gamma \nabla f(x_k)\right).$$

1. **Fixed-point characterization: Show that $x^* \in \mathbb{R}^n$ is a global minimizer of F if and only if x^* is a fixed point of the proximal gradient mapping, i.e.,**

$$x^* = \text{prox}_{\gamma g}\left(x^* - \gamma \nabla f(x^*)\right).$$

2. **Majorization-Minimization viewpoint: Show that the iteration x_{k+1} is the exact minimizer of a quadratic surrogate function:**

$$x_{k+1} = \arg\min_{u\in\mathbb{R}^n} \left\{ f(x_k) + \langle \nabla f(x_k), u - x_k\rangle + \frac{1}{2\gamma}\|u - x_k\|_2^2 + g(u) \right\}.$$

3. **Application to LASSO (ISTA): Consider the LASSO problem:**

$$\min_{x\in\mathbb{R}^n} \frac{1}{2}\|Ax - b\|_2^2 + \lambda\|x\|_1,$$

where $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and $\lambda > 0$. Identify f and g, compute $\nabla f(x)$, and derive the explicit update rule of Proximal Gradient Descent using the component-wise soft-thresholding operator $\mathcal{S}_\tau(t) = \text{sign}(t)\max(|t| - \tau, 0)$.

****

Solution of the exercise 4.13 1. Fixed-point characterization of optimality:
Since f is convex differentiable and g is convex proper l.s.c., Fermat's rule states that x^* minimizes $F = f + g$ if and only if:

$$0 \in \nabla f(x^*) + \partial g(x^*).$$

Multiplying this inclusion by $\gamma > 0$ preserves equivalence:

$$-\gamma\nabla f(x^*) \in \gamma\partial g(x^*).$$

Adding $x^* - x^*$ to the left-hand side yields:

$$(x^* - \gamma\nabla f(x^*)) - x^* \in \gamma\partial g(x^*).$$

Recall that for any $y \in \mathbb{R}^n$, the proximal operator $p = \text{prox}_{\gamma g}(y)$ is characterized by:

$$y - p \in \gamma\partial g(p).$$

Setting $y = x^* - \gamma\nabla f(x^*)$ and identifying $p = x^*$, we obtain:

$$(x^* - \gamma\nabla f(x^*)) - x^* \in \gamma\partial g(x^*) \iff x^* = \text{prox}_{\gamma g}\left(x^* - \gamma\nabla f(x^*)\right).$$

Thus, the minimizers of $f + g$ are precisely the fixed points of the proximal gradient operator.

2. Majorization-Minimization formulation:
By definition of the proximal operator:

$$x_{k+1} = \text{prox}_{\gamma g}\left(x_k - \gamma\nabla f(x_k)\right) = \arg\min_{u\in\mathbb{R}^n} \left\{ g(u) + \frac{1}{2\gamma}\left\|u - (x_k - \gamma\nabla f(x_k))\right\|_2^2 \right\}.$$

Expanding the squared Euclidean norm:

$$\begin{aligned}\frac{1}{2\gamma}\left\|(u - x_k) + \gamma\nabla f(x_k)\right\|_2^2 &= \frac{1}{2\gamma}\|u - x_k\|_2^2 + \frac{1}{\gamma}\langle\gamma\nabla f(x_k), u - x_k\rangle + \frac{\gamma^2}{2\gamma}\|\nabla f(x_k)\|_2^2 \\ &= \langle\nabla f(x_k), u - x_k\rangle + \frac{1}{2\gamma}\|u - x_k\|_2^2 + \frac{\gamma}{2}\|\nabla f(x_k)\|_2^2.\end{aligned}$$

The last term $\frac{\gamma}{2}\|\nabla f(x_k)\|_2^2$ and the constant $f(x_k)$ do not depend on the optimization variable u. Adding them to the objective leaves the argmin unchanged:

$$x_{k+1} = \arg\min_{u\in\mathbb{R}^n} \left\{ f(x_k) + \langle \nabla f(x_k), u - x_k\rangle + \frac{1}{2\gamma}\|u - x_k\|_2^2 + g(u) \right\}.$$

This shows that proximal gradient descent approximates f by its first-order Taylor expansion regularized by a quadratic proximity penalty, and minimizes this surrogate combined with the non-smooth term g.

3. Application to LASSO (ISTA):

We identify the smooth and non-smooth components:

$$f(x) = \frac{1}{2}\|Ax - b\|_2^2 \quad \text{and} \quad g(x) = \lambda\|x\|_1.$$

The gradient of f is:

$$\nabla f(x) = A^\top(Ax - b).$$

The proximal gradient update is:

$$x_{k+1} = \text{prox}_{\gamma\lambda\|\cdot\|_1}\left(x_k - \gamma A^\top(Ax_k - b)\right).$$

Since the ℓ_1-norm is separable ($g(x) = \sum_{i=1}^n \lambda|x_i|$), its proximal operator decomposes coordinate-wise:

$$\left[\text{prox}_{\gamma\lambda\|\cdot\|_1}(z)\right]_i = \text{prox}_{\gamma\lambda|\cdot|}(z_i) = \mathcal{S}_{\gamma\lambda}(z_i).$$

Letting $z_k = x_k - \gamma A^\top(Ax_k - b)$ denote the forward gradient step, each coordinate is updated via:

$$(x_{k+1})_i = \mathcal{S}_{\gamma\lambda}\left((x_k)_i - \gamma\left[A^\top(Ax_k - b)\right]_i\right) = \text{sign}((z_k)_i)\max\left(|(z_k)_i| - \gamma\lambda,\, 0\right).$$

This algorithm is known as the **Iterative Soft-Thresholding Algorithm (ISTA)**. It converges to a global minimizer whenever the step size satisfies $0 < \gamma \leq \frac{1}{\|A\|_2^2}$, where $\|A\|_2^2$ is the Lipschitz constant of ∇f.

****

Exercise 4.14 (The Orthogonal Procrustes Problem (Projection onto $\mathcal{O}(n)$))

Let $\mathcal{O}(n) = \{Q \in \mathbb{R}^{n\times n} \mid Q^\top Q = I_n\}$ denote the orthogonal group, and let $M \in \mathbb{R}^{n\times n}$ be an arbitrary matrix. Consider the projection problem onto the non-convex set $\mathcal{O}(n)$ under the Frobenius norm:

$$\min_{Q\in\mathcal{O}(n)} \|Q - M\|_F^2,$$

where $\|A\|_F = \sqrt{\text{Tr}(A^\top A)}$.

1. Show that this minimization problem is equivalent to:

$$\max_{Q\in\mathcal{O}(n)} \text{Tr}(Q^\top M).$$

2. **Let $M = U\Sigma V^\top$ be a Singular Value Decomposition (SVD) of M, with $U, V \in \mathcal{O}(n)$ and $\Sigma = \mathrm{diag}(\sigma_1, \dots, \sigma_n)$ where $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Show that:**

$$\max_{Q \in \mathcal{O}(n)} \mathrm{Tr}(Q^\top M) = \sum_{i=1}^{n} \sigma_i,$$

and verify that $Q^* = UV^\top$ is a global minimizer.

3. **Determine a necessary and sufficient condition on M for the minimizer Q^* to be unique.**

****

Solution of the exercise 4.14 1. Equivalence to trace maximization:
Expanding the squared Frobenius norm:

$$\begin{aligned}\|Q - M\|_F^2 &= \mathrm{Tr}\left((Q-M)^\top (Q-M)\right) \\ &= \mathrm{Tr}(Q^\top Q) - 2\,\mathrm{Tr}(Q^\top M) + \mathrm{Tr}(M^\top M).\end{aligned}$$

Since $Q \in \mathcal{O}(n)$, we have $Q^\top Q = I_n$, so $\mathrm{Tr}(Q^\top Q) = \mathrm{Tr}(I_n) = n$. Thus:

$$\|Q - M\|_F^2 = n + \|M\|_F^2 - 2\,\mathrm{Tr}(Q^\top M).$$

Because $n + \|M\|_F^2$ is independent of Q, minimizing $\|Q - M\|_F^2$ over $\mathcal{O}(n)$ is equivalent to:

$$\max_{Q \in \mathcal{O}(n)} \mathrm{Tr}(Q^\top M).$$

2. SVD-based derivation of the global maximizer:
Substitute the SVD $M = U\Sigma V^\top$ into the trace:

$$\mathrm{Tr}(Q^\top M) = \mathrm{Tr}(Q^\top U \Sigma V^\top) = \mathrm{Tr}(V^\top Q^\top U \Sigma).$$

Define the matrix $Z := V^\top Q^\top U$. Because V, Q, and U are all orthogonal, Z is also an orthogonal matrix:

$$Z^\top Z = (U^\top Q V)(V^\top Q^\top U) = U^\top Q I_n Q^\top U = U^\top I_n U = I_n.$$

The trace becomes:

$$\mathrm{Tr}(Z\Sigma) = \sum_{i=1}^{n} Z_{ii}\sigma_i.$$

Since $Z \in \mathcal{O}(n)$, each column has unit Euclidean norm:

$$\sum_{j=1}^{n} Z_{ji}^2 = 1 \implies |Z_{ii}| \leq 1 \quad \text{for all } i \in \{1, \dots, n\}.$$

Because the singular values satisfy $\sigma_i \geq 0$, we establish the upper bound:

$$\mathrm{Tr}(Z\Sigma) = \sum_{i=1}^{n} Z_{ii}\sigma_i \leq \sum_{i=1}^{n} 1 \cdot \sigma_i = \sum_{i=1}^{n} \sigma_i.$$

This bound is attained when $Z_{ii} = 1$ for all i. Choosing $Z = I_n$ achieves this maximum. Solving for Q:

$$V^\top Q^\top U = I_n \iff Q^\top = VU^\top \iff Q^* = UV^\top.$$

Since $(Q^*)^\top Q^* = VU^\top UV^\top = VI_nV^\top = I_n$, $Q^* \in \mathcal{O}(n)$ is indeed a feasible point and hence a global minimizer.

3. Condition for uniqueness:

The maximum value $\sum_{i=1}^n \sigma_i$ is attained if and only if:

$$\sum_{i=1}^{n}(1 - Z_{ii})\sigma_i = 0.$$

Since $1 - Z_{ii} \geq 0$ and $\sigma_i \geq 0$, each term must be zero:

$$(1 - Z_{ii})\sigma_i = 0 \quad \text{for all } i \in \{1, \ldots, n\}.$$

- If M is non-singular ($\det(M) \neq 0$), then $\sigma_i > 0$ for all $i \in \{1, \ldots, n\}$. This forces $Z_{ii} = 1$ for every i. Since $\sum_{j=1}^n Z_{ji}^2 = 1$ and $Z_{ii} = 1$, all off-diagonal entries in the i-th column must be zero ($Z_{ji} = 0$ for $j \neq i$). Hence $Z = I_n$ is uniquely determined, and $Q^* = UV^\top$ is unique.
- If M is singular ($\det(M) = 0$), there exists at least one i such that $\sigma_i = 0$. The corresponding entry Z_{ii} can be chosen strictly less than 1 without altering the sum (for example, taking $Z_{nn} = -1$), yielding multiple orthogonal matrices that achieve the exact same minimum.

Thus, the projection is unique if and only if M is invertible ($\det(M) \neq 0$).

****

Exercise 4.15 (The simplex method)

Solve the following linear program by the simplex method:

$$\min \left(-5x_1 - 2x_2 - 3x_3 : \begin{array}{rl} x_1 + x_2 & \leq 5 \\ 2x_1 - x_2 + x_3 & \leq 10 \\ x_2 - x_3 & \leq 4 \\ x_1, x_2, x_3 \geq 0 & \end{array} \right)$$

****

Solution of the exercise 4.15

z	x_1	x_2	x_3	x_4	x_5	x_6	b
1	5	2	3	0	0	0	0
0	(1)	1	0	1	0	0	5
0	2	−1	1	0	1	0	10
0	0	1	−1	0	0	1	4
1	0	−3	3	−5	0	0	−25
0	1	1	0	1	0	0	5
0	0	−3	(1)	−2	1	0	0
0	0	1	−1	0	0	1	4
1	0	6	0	1	−3	0	−25
0	1	1	0	(1)	0	0	5
0	0	−3	1	−2	1	0	0
0	0	−2	0	−2	1	1	4
1	−1	5	0	0	−3	0	−30
0	1	(1)	0	1	0	0	5
0	2	−1	1	0	1	0	10
0	2	0	0	0	1	1	14
1	−6	0	0	−5	−3	0	−55
0	1	1	0	1	0	0	5
0	3	0	1	1	1	0	15
0	2	0	0	0	1	1	14

The minimum equals -55, and it is attained at $x_1 = 0$, $x_2 = 5$ and $x_3 = 15$.

****

Exercise 4.16 (Computation of Fenchel Conjugates)

Let $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be a proper function (i.e. not always equal to $+\infty$). The Fenchel conjugate (or Legendre-Fenchel transform) $f^* : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is defined by:

$$f^*(y) := \sup_{x \in \mathbb{R}^n} \{\langle y, x\rangle - f(x)\}.$$

1. **Strictly convex quadratic: Let $A \in \mathcal{S}_n^{++}$ (a symmetric positive definite matrix) and $b \in \mathbb{R}^n$. Compute the conjugate of:**

$$f(x) = \frac{1}{2}\langle x, Ax\rangle - \langle b, x\rangle.$$

2. **Norm function: Let $\|\cdot\|$ be an arbitrary norm on $\mathbb{R}^n$. Recall that its dual norm is defined by $\|y\|_* := \sup_{\|x\|\le 1}\langle y, x\rangle$. Show that the conjugate of $f(x) = \|x\|$ is the indicator function of the dual unit ball:**

$$f^*(y) = \iota_{\mathcal{B}_*}(y) = \begin{cases} 0 & \text{if } \|y\|_* \le 1, \\ +\infty & \text{if } \|y\|_* > 1, \end{cases}$$

where $\mathcal{B}_* = \{y \in \mathbb{R}^n \mid \|y\|_* \le 1\}$.

3. **Negative entropy: Let $f : \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ be the univariate negative entropy function defined by:**

$$f(x) = \begin{cases} x\ln(x) & \textbf{if } x > 0, \\ 0 & \textbf{if } x = 0, \\ +\infty & \textbf{if } x < 0. \end{cases}$$

Compute $f^*(y)$ for all $y \in \mathbb{R}$.

****

Solution of the exercise 4.16 1. Conjugate of the strictly convex quadratic function:
For a fixed $y \in \mathbb{R}^n$, we consider the objective:

$$\phi_y(x) := \langle y, x\rangle - f(x) = \langle y + b, x\rangle - \frac{1}{2}\langle x, Ax\rangle.$$

Since A is symmetric positive definite ($A \in \mathcal{S}_n^{++}$), the Hessian of ϕ_y is $\nabla^2\phi_y(x) = -A \prec 0$. Hence, ϕ_y is strictly concave and attains its unique global maximum at its stationary point:

$$\nabla\phi_y(x) = 0 \iff (y + b) - Ax = 0 \iff x^* = A^{-1}(y + b).$$

Substituting x^* back into ϕ_y:

$$\begin{aligned} f^*(y) = \phi_y(x^*) &= \langle y + b, A^{-1}(y + b)\rangle - \frac{1}{2}\langle A^{-1}(y + b), AA^{-1}(y + b)\rangle \\ &= \langle y + b, A^{-1}(y + b)\rangle - \frac{1}{2}\langle y + b, A^{-1}(y + b)\rangle \\ &= \frac{1}{2}\langle y + b, A^{-1}(y + b)\rangle. \end{aligned}$$

In particular, when $b = 0$ and $A = I_n$ ($f(x) = \frac{1}{2}\|x\|_2^2$), we recover $f^*(y) = \frac{1}{2}\|y\|_2^2$.

2. Conjugate of a general norm:
By definition:

$$f^*(y) = \sup_{x\in\mathbb{R}^n} \{\langle y, x\rangle - \|x\|\}.$$

Recall that the dual norm satisfies the generalized Cauchy-Schwarz inequality:

$$\langle y, x\rangle \le \|y\|_*\|x\| \quad \text{for all } x, y \in \mathbb{R}^n.$$

We evaluate the supremum in two cases:

- **Case $\|y\|_* \le 1$:**
  For every $x \in \mathbb{R}^n$:

$$\langle y, x\rangle - \|x\| \le \|y\|_*\|x\| - \|x\| = (\|y\|_* - 1)\|x\| \le 0.$$

  Choosing $x = 0$ gives $\langle y, 0\rangle - \|0\| = 0$. Since 0 is an upper bound and is attained at $x = 0$, the supremum is 0:

$$f^*(y) = 0.$$

- **Case $\|y\|_* > 1$:**
  By definition of the dual norm as a supremum on the unit sphere, there exists a vector $\bar{x} \in \mathbb{R}^n$ with $\|\bar{x}\| = 1$ such that $\langle y, \bar{x} \rangle = \|y\|_*$.
  Consider the ray $x_t = t\bar{x}$ for $t > 0$. Evaluating the objective along this direction:

$$\langle y, x_t \rangle - \|x_t\| = t\langle y, \bar{x} \rangle - t\|\bar{x}\| = t(\|y\|_* - 1).$$

  Since $\|y\|_* > 1$, the quantity $\|y\|_* - 1$ is strictly positive. Letting $t \to +\infty$:

$$\lim_{t \to +\infty} t(\|y\|_* - 1) = +\infty.$$

  Therefore, the supremum is $+\infty$.

Combining both cases gives the indicator of the closed dual unit ball:

$$f^*(y) = \begin{cases} 0 & \text{if } \|y\|_* \leq 1, \\ +\infty & \text{if } \|y\|_* > 1. \end{cases}$$

3. Conjugate of the negative entropy:

Since $f(x) = +\infty$ for $x < 0$, the effective domain is $\mathrm{dom}(f) = [0, +\infty)$. For any fixed $y \in \mathbb{R}$:

$$f^*(y) = \sup_{x \geq 0} \{yx - f(x)\} = \max \left\{ 0, \ \sup_{x > 0} \{yx - x \ln(x)\} \right\},$$

where the candidate 0 corresponds to $x = 0$ ($y \cdot 0 - f(0) = 0$).
Define $g(x) = yx - x\ln(x)$ on $(0, +\infty)$. Taking derivatives:

$$g'(x) = y - (\ln(x) + 1) = y - \ln(x) - 1,$$

$$g''(x) = -\frac{1}{x} < 0 \quad \text{for all } x > 0.$$

Because g is strictly concave on $(0, +\infty)$, its global maximum on $(0, +\infty)$ is attained at the unique critical point:

$$g'(x) = 0 \iff \ln(x) = y - 1 \iff x^* = e^{y-1} > 0.$$

Evaluating g at this stationary point:

$$g(x^*) = ye^{y-1} - e^{y-1}\ln(e^{y-1}) = ye^{y-1} - e^{y-1}(y-1) = e^{y-1}.$$

Since $e^{y-1} > 0 = g(0)$ for all $y \in \mathbb{R}$, the maximum over $[0, +\infty)$ is attained at $x^* > 0$. Thus:

$$f^*(y) = e^{y-1}.$$

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Exercise 4.17 (Biconjugate Function and Convex Envelope)

Let $f : \mathbb{R} \to \mathbb{R}$ be the double-well function defined by:

$$f(x) = (|x| - 1)^2.$$

The Fenchel conjugate f^* and the biconjugate f^{**} are defined respectively by:

$$f^*(y) := \sup_{x \in \mathbb{R}} \{yx - f(x)\}, \quad f^{**}(x) := \sup_{y \in \mathbb{R}} \{xy - f^*(y)\}.$$

1. **Show that f is not convex.**
2. **Compute the first Fenchel conjugate $f^*(y)$ for all $y \in \mathbb{R}$.**
3. **Compute the Fenchel biconjugate $f^{**}(x)$ for all $x \in \mathbb{R}$.**
4. **Show that $f^{**}(x) \leq f(x)$ for all $x \in \mathbb{R}$, and interpret f^{**} geometrically in terms of the convex envelope $\overline{\text{conv}}(f)$.**

****

Solution of the exercise 4.17 1. Non-convexity of f:
We evaluate f at the points $x_1 = -1$, $x_2 = 1$, and their midpoint $x_0 = \frac{x_1+x_2}{2} = 0$:

$$f(-1) = (|-1|-1)^2 = 0, \quad f(1) = (|1|-1)^2 = 0, \quad f(0) = (|0|-1)^2 = 1.$$

If f were convex, Jensen's inequality would require:

$$f(0) \leq \frac{1}{2}f(-1) + \frac{1}{2}f(1) = \frac{1}{2}(0) + \frac{1}{2}(0) = 0.$$

Since $f(0) = 1 > 0$, the inequality fails, proving that f is not convex.

2. First conjugate $f^*(y)$:
By definition:

$$f^*(y) = \sup_{x\in\mathbb{R}} \left\{yx - (|x|-1)^2\right\}.$$

Since $f(-x) = (|-x|-1)^2 = f(x)$, f is an even function. Therefore:

$$f^*(-y) = \sup_{x\in\mathbb{R}}\{-yx - f(x)\} = \sup_{u\in\mathbb{R}}\{yu - f(-u)\} = \sup_{u\in\mathbb{R}}\{yu - f(u)\} = f^*(y).$$

Hence f^* is also even. We can assume $y \geq 0$ and partition the supremum into $x \geq 0$ and $x \leq 0$:

- **On $x \geq 0$:** The objective becomes:

  $$\phi(x) := yx - (x-1)^2 = -x^2 + (y+2)x - 1.$$

  This is a concave quadratic whose unconstrained global maximizer is:

  $$x^* = \frac{y+2}{2} = 1 + \frac{y}{2}.$$

  Since $y \geq 0$, we have $x^* \geq 1 > 0$, which belongs to the domain $[0, +\infty)$. The maximum value is:

  $$\phi(x^*) = -\left(1+\frac{y}{2}\right)^2 + (y+2)\left(1+\frac{y}{2}\right) - 1 = \left(1+\frac{y}{2}\right)^2 - 1 = y + \frac{y^2}{4}.$$

- **On $x \leq 0$:** Setting $u = -x \geq 0$:

  $$yx - (|x|-1)^2 = -yu - (u-1)^2 = -u^2 - (y-2)u - 1.$$

  For $u \geq 0$ and $y \geq 0$, this function attains values strictly upper-bounded by $y + \frac{y^2}{4}$ (with equality only at $y = 0$).

Thus, for $y \geq 0$, $f^*(y) = y + \frac{y^2}{4}$. Using the symmetry $f^*(-y) = f^*(y)$, we obtain for all $y \in \mathbb{R}$:

$$f^*(y) = |y| + \frac{y^2}{4}.$$

3. Biconjugate $f^{**}(x)$:

By definition:

$$f^{**}(x) = \sup_{y \in \mathbb{R}} \left\{ xy - |y| - \frac{y^2}{4} \right\}.$$

Since $f^*(y)$ is even, $f^{**}(x)$ is also an even function ($f^{**}(-x) = f^{**}(x)$). We assume $x \geq 0$:

- **For $y \geq 0$:** The expression is:

$$\psi(y) := (x-1)y - \frac{1}{4}y^2.$$

  The unconstrained stationary point of ψ is $y^* = 2(x-1)$:

  - If $0 \leq x \leq 1$: We have $x - 1 \leq 0$, so $y^* \leq 0$. Since ψ is strictly concave, its maximum over $y \in [0, +\infty)$ is attained at the boundary $y = 0$:

$$\sup_{y \geq 0} \psi(y) = \psi(0) = 0.$$

  - If $x > 1$: We have $x - 1 > 0$, so $y^* = 2(x-1) > 0$ is feasible. Evaluating at y^*:

$$\psi(y^*) = 2(x-1)^2 - \frac{1}{4}\left(4(x-1)^2\right) = (x-1)^2.$$

- **For $y < 0$:** Setting $t = -y > 0$:

$$xy - |y| - \frac{y^2}{4} = -(x+1)t - \frac{1}{4}t^2 < 0 \quad \text{(since } x \geq 0 \text{ and } t > 0\text{)}.$$

  This candidate is strictly negative and never achieves the supremum.

Therefore, for $x \geq 0$:

$$f^{**}(x) = \begin{cases} 0 & \text{if } 0 \leq x \leq 1, \\ (x-1)^2 & \text{if } x > 1. \end{cases}$$

By symmetry, for all $x \in \mathbb{R}$:

$$f^{**}(x) = \begin{cases} 0 & \text{if } |x| \leq 1, \\ (|x|-1)^2 & \text{if } |x| > 1. \end{cases}$$

4. Comparison and convex envelope interpretation:

- **Comparison:**

  - For $|x| > 1$: $f^{**}(x) = (|x|-1)^2 = f(x)$.
  - For $|x| \leq 1$: $f^{**}(x) = 0 \leq (|x|-1)^2 = f(x)$, with strict inequality on $(-1, 1)$ (e.g., at $x = 0$, $f^{**}(0) = 0 < 1 = f(0)$).

  Hence $f^{**}(x) \leq f(x)$ holds for all $x \in \mathbb{R}$.

- **Convex envelope:** According to the Fenchel-Moreau theorem, the biconjugate f^{**} is the greatest convex and lower semicontinuous function minorizing f:

$$f^{**} = \overline{\text{conv}}(f).$$

Geometrically, the epigraph of f^{**} is the closed convex hull of the epigraph of f. Over the interval $[-1, 1]$, f exhibits a non-convex hump; the biconjugate operation bridges this non-convex gap with the constant affine segment $f^{**}(x) = 0$, which is the supporting tangent line connecting the two global minima at $x = -1$ and $x = 1$.

5 Matrix analysis

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Exercise 5.1 (Frobenius Inner Product and Rank-One Matrices)

Let $\mathcal{M}_{m,n}(\mathbb{R})$ be the space of $m \times n$ real matrices equipped with the Frobenius inner product defined by:

$$\langle A, B \rangle = \text{Tr}(A^\top B).$$

1. **Show that $\langle A, B \rangle = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} B_{ij} = \text{vec}(A)^\top \text{vec}(B)$, where $\text{vec}(A) \in \mathbb{R}^{mn}$ stacks the columns of A.**

2. **Let $u, x \in \mathbb{R}^m$ and $v, y \in \mathbb{R}^n$. Prove that the inner product of the rank-one matrices $uv^\top$ and $xy^\top$ satisfies:**

$$\langle uv^\top, xy^\top \rangle = (u^\top x)(v^\top y).$$

3. **Let $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $v = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, $x = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$, and $y = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$. Compute the matrices $A = uv^\top$ and $B = xy^\top$, then calculate $\langle A, B \rangle$ both directly via the sum of entrywise products and using the formula established in question 2.**

****

Solution of the exercise 5.1 1. Component-wise formulation and vectorization:
By definition of matrix multiplication, the (j, j)-th entry of the $n \times n$ matrix $A^\top B$ is:

$$(A^\top B)_{jj} = \sum_{i=1}^{m} (A^\top)_{ji} B_{ij} = \sum_{i=1}^{m} A_{ij} B_{ij}.$$

Taking the trace yields:

$$\langle A, B \rangle = \text{Tr}(A^\top B) = \sum_{j=1}^{n} (A^\top B)_{jj} = \sum_{j=1}^{n} \sum_{i=1}^{m} A_{ij} B_{ij} = \sum_{i=1}^{m} \sum_{j=1}^{n} A_{ij} B_{ij}.$$

Letting $\text{vec}(M) \in \mathbb{R}^{mn}$ denote the vector formed by unrolling the matrix entries, this double sum is precisely the standard Euclidean dot product:

$$\langle A, B \rangle = \text{vec}(A)^\top \text{vec}(B).$$

2. Inner product of rank-one matrices:
Using the cyclic property of the trace $\mathrm{Tr}(MN) = \mathrm{Tr}(NM)$ and the transpose identity $(uv^\top)^\top = vu^\top$:

$$\langle uv^\top, xy^\top \rangle = \mathrm{Tr}\left((uv^\top)^\top (xy^\top)\right) = \mathrm{Tr}(vu^\top xy^\top).$$

Since $u^\top x \in \mathbb{R}$ is a scalar, it factors out of the trace:

$$\mathrm{Tr}(v(u^\top x)y^\top) = (u^\top x)\,\mathrm{Tr}(vy^\top).$$

Applying the cyclic property to $\mathrm{Tr}(vy^\top)$ gives $\mathrm{Tr}(y^\top v) = y^\top v = v^\top y$. Thus:

$$\langle uv^\top, xy^\top \rangle = (u^\top x)(v^\top y).$$

3. Concrete numerical computation:
We compute the rank-one outer products:

$$A = uv^\top = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -1 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 6 & -2 \end{pmatrix},$$

$$B = xy^\top = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \begin{pmatrix} 2 & 4 \end{pmatrix} = \begin{pmatrix} -2 & -4 \\ 2 & 4 \end{pmatrix}.$$

- **Direct component-wise calculation:**

$$\langle A, B \rangle = 3(-2) + (-1)(-4) + 6(2) + (-2)(4) = -6 + 4 + 12 - 8 = 2.$$

- **Using the rank-one factorized formula:**

$$u^\top x = 1(-1) + 2(1) = 1, \quad v^\top y = 3(2) + (-1)(4) = 6 - 4 = 2.$$

$$\langle A, B \rangle = (u^\top x)(v^\top y) = 1 \times 2 = 2.$$

Both methods yield the same result.

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Exercise 5.2 (Orthogonal Decomposition and Projection onto Symmetric Matrices)

Let $\mathcal{M}_n(\mathbb{R})$ be the space of square real matrices endowed with the Frobenius inner product $\langle X, Y \rangle = \mathrm{Tr}(X^\top Y)$. Consider the subspaces of symmetric and skew-symmetric matrices:

$$\mathcal{S}_n = \{M \in \mathcal{M}_n(\mathbb{R}) \mid M^\top = M\}, \quad \mathcal{A}_n = \{M \in \mathcal{M}_n(\mathbb{R}) \mid M^\top = -M\}.$$

1. **Show that $\mathcal{S}_n$ and $\mathcal{A}_n$ are orthogonal with respect to the Frobenius inner product, i.e.,**

$$\forall S \in \mathcal{S}_n, \quad \forall A \in \mathcal{A}_n, \quad \langle S, A \rangle = 0.$$

2. **Deduce that $\mathcal{M}_n(\mathbb{R}) = \mathcal{S}_n \overset{\perp}{\oplus} \mathcal{A}_n$, and give the closed-form expression for the orthogonal projection $\Pi_{\mathcal{S}_n}(M)$ of an arbitrary matrix $M \in \mathcal{M}_n(\mathbb{R})$ onto $\mathcal{S}_n$.**

3. Consider the matrix:

$$M = \begin{pmatrix} 2 & 4 \\ -2 & 6 \end{pmatrix}.$$

Compute $\Pi_{\mathcal{S}_2}(M)$ and $\Pi_{\mathcal{A}_2}(M)$, verify that $\langle \Pi_{\mathcal{S}_2}(M), \Pi_{\mathcal{A}_2}(M) \rangle = 0$, and compute the Frobenius distance from M to $\mathcal{S}_2$:

$$\mathrm{dist}_F(M, \mathcal{S}_2) = \min_{S \in \mathcal{S}_2} \|M - S\|_F.$$

****

Solution of the exercise 5.2 1. Orthogonality of $\mathcal{S}_n$ and $\mathcal{A}_n$:
Let $S \in \mathcal{S}_n$ (so $S^\top = S$) and $A \in \mathcal{A}_n$ (so $A^\top = -A$). Using the trace identity $\mathrm{Tr}(Z^\top) = \mathrm{Tr}(Z)$:

$$\langle S, A \rangle = \mathrm{Tr}(S^\top A) = \mathrm{Tr}(SA).$$

Transposing inside the trace:

$$\mathrm{Tr}(SA) = \mathrm{Tr}\left((SA)^\top\right) = \mathrm{Tr}(A^\top S^\top) = \mathrm{Tr}((-A)S) = -\mathrm{Tr}(AS).$$

By the cyclic property $\mathrm{Tr}(AS) = \mathrm{Tr}(SA)$, hence:

$$\mathrm{Tr}(SA) = -\mathrm{Tr}(SA) \implies 2\,\mathrm{Tr}(SA) = 0 \implies \langle S, A \rangle = 0.$$

Thus, $\mathcal{S}_n \perp \mathcal{A}_n$.

2. Orthogonal direct sum and projection:
Every matrix $M \in \mathcal{M}_n(\mathbb{R})$ can be decomposed algebraically as:

$$M = \frac{M + M^\top}{2} + \frac{M - M^\top}{2}.$$

Setting $S = \frac{M+M^\top}{2}$ and $A = \frac{M-M^\top}{2}$:

$$S^\top = \frac{M^\top + M}{2} = S \in \mathcal{S}_n, \quad A^\top = \frac{M^\top - M}{2} = -A \in \mathcal{A}_n.$$

Because $\mathcal{S}_n \cap \mathcal{A}_n = \{0\}$ and $\dim(\mathcal{S}_n) + \dim(\mathcal{A}_n) = \frac{n(n+1)}{2} + \frac{n(n-1)}{2} = n^2 = \dim(\mathcal{M}_n(\mathbb{R}))$, and because $\mathcal{S}_n \perp \mathcal{A}_n$, the direct sum is orthogonal:

$$\mathcal{M}_n(\mathbb{R}) = \mathcal{S}_n \overset{\perp}{\oplus} \mathcal{A}_n.$$

The orthogonal projection of M onto $\mathcal{S}_n$ is the symmetric component:

$$\Pi_{\mathcal{S}_n}(M) = \frac{M + M^\top}{2}.$$

3. Application to the given matrix:
For $M = \begin{pmatrix} 2 & 4 \\ -2 & 6 \end{pmatrix}$, we have $M^\top = \begin{pmatrix} 2 & -2 \\ 4 & 6 \end{pmatrix}$.

- Symmetric projection:

$$\Pi_{\mathcal{S}_2}(M) = \frac{1}{2}\left[\begin{pmatrix} 2 & 4 \\ -2 & 6 \end{pmatrix} + \begin{pmatrix} 2 & -2 \\ 4 & 6 \end{pmatrix}\right] = \begin{pmatrix} 2 & 1 \\ 1 & 6 \end{pmatrix}.$$

- Skew-symmetric projection:

$$\Pi_{\mathcal{A}_2}(M) = \frac{1}{2}\left[\begin{pmatrix} 2 & 4 \\ -2 & 6 \end{pmatrix} - \begin{pmatrix} 2 & -2 \\ 4 & 6 \end{pmatrix}\right] = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}.$$

- Orthogonality check:

$$\langle \Pi_{\mathcal{S}_2}(M), \Pi_{\mathcal{A}_2}(M) \rangle = 2(0) + 1(3) + 1(-3) + 6(0) = 3 - 3 = 0.$$

- Distance to $\mathcal{S}_2$: By the Pythagorean theorem, the distance to the subspace $\mathcal{S}_2$ is the norm of the orthogonal component $\Pi_{\mathcal{A}_2}(M)$:

$$\text{dist}_F(M, \mathcal{S}_2) = \|M - \Pi_{\mathcal{S}_2}(M)\|_F = \|\Pi_{\mathcal{A}_2}(M)\|_F = \sqrt{0^2 + 3^2 + (-3)^2 + 0^2} = \sqrt{18} = 3\sqrt{2}.$$

****

Exercise 5.3 (Unitary Invariance and Singular Value Representation of the Frobenius Norm)
Let $\mathcal{M}_{m,n}(\mathbb{R})$ denote the space of $m \times n$ real matrices equipped with the Frobenius norm:

$$\|A\|_F = \sqrt{\text{Tr}(A^\top A)} = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}^2}.$$

1. **Unitary invariance: Let $U \in \mathcal{O}(m)$ and $V \in \mathcal{O}(n)$ be orthogonal matrices (i.e., $U^\top U = I_m$ and $V^\top V = I_n$). Prove that:**

$$\|UAV^\top\|_F = \|A\|_F.$$

2. **Singular values representation: Let $A = U\Sigma V^\top$ be a Singular Value Decomposition (SVD) of A, where $\Sigma \in \mathcal{M}_{m,n}(\mathbb{R})$ has the singular values $\sigma_1 \geq \cdots \geq \sigma_r > 0$ on its main diagonal ($r = \text{rank}(A)$). Deduce that:**

$$\|A\|_F = \sqrt{\sum_{i=1}^{\min(m,n)} \sigma_i(A)^2}.$$

3. **Submultiplicativity: Show that for any $A \in \mathcal{M}_{m,n}(\mathbb{R})$ and $B \in \mathcal{M}_{n,p}(\mathbb{R})$:**

$$\|AB\|_F \leq \|A\|_F \|B\|_F.$$

4. **Concrete computation: Consider the rank-one matrix:**

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

(a) Compute $\|A\|_F$ directly from its entries.

(b) Determine the singular values of A and verify the identity established in question 2.

****

Solution of the exercise 5.3 1. Unitary invariance:
By definition of the Frobenius norm via the trace:

$$\|UAV^\top\|_F^2 = \mathrm{Tr}\left((UAV^\top)^\top(UAV^\top)\right) = \mathrm{Tr}\left(VA^\top U^\top UAV^\top\right).$$

Since $U \in \mathcal{O}(m)$, we have $U^\top U = I_m$, which simplifies the product to:

$$\mathrm{Tr}\left(VA^\top I_m AV^\top\right) = \mathrm{Tr}\left(VA^\top AV^\top\right).$$

Applying the cyclic permutation property of the trace ($\mathrm{Tr}(XY) = \mathrm{Tr}(YX)$ with $X = V$ and $Y = A^\top AV^\top$):

$$\mathrm{Tr}\left(V(A^\top AV^\top)\right) = \mathrm{Tr}\left((A^\top AV^\top)V\right) = \mathrm{Tr}\left(A^\top A(V^\top V)\right).$$

Since $V \in \mathcal{O}(n)$, we have $V^\top V = I_n$, hence:

$$\|UAV^\top\|_F^2 = \mathrm{Tr}(A^\top A) = \|A\|_F^2.$$

Taking the square root yields $\|UAV^\top\|_F = \|A\|_F$.

2. Relation with singular values:
Let $A = U\Sigma V^\top$ be the SVD of A, with $U \in \mathcal{O}(m)$ and $V \in \mathcal{O}(n)$. By unitary invariance:

$$\|A\|_F = \|U\Sigma V^\top\|_F = \|\Sigma\|_F.$$

The matrix $\Sigma \in \mathcal{M}_{m,n}(\mathbb{R})$ contains the singular values $\sigma_i(A)$ on its diagonal and zero elsewhere:

$$\|\Sigma\|_F^2 = \sum_{i=1}^{m}\sum_{j=1}^{n}\Sigma_{ij}^2 = \sum_{i=1}^{\min(m,n)}\Sigma_{ii}^2 = \sum_{i=1}^{\min(m,n)}\sigma_i(A)^2.$$

Taking the square root gives:

$$\|A\|_F = \sqrt{\sum_{i=1}^{\min(m,n)}\sigma_i(A)^2}.$$

This shows that the Frobenius norm coincides with the ℓ_2-norm of the vector of singular values (the Schatten-2 norm).

3. Submultiplicativity:
Let $a_i^\top \in \mathbb{R}^{1\times n}$ denote the i-th row of A, and let $b_j \in \mathbb{R}^{n\times 1}$ denote the j-th column of B. The (i,j)-th entry of the product is $(AB)_{ij} = a_i^\top b_j = \langle a_i, b_j\rangle$. Writing the Frobenius norm entrywise:

$$\|AB\|_F^2 = \sum_{i=1}^{m}\sum_{j=1}^{p}|(AB)_{ij}|^2 = \sum_{i=1}^{m}\sum_{j=1}^{p}|\langle a_i, b_j\rangle|^2.$$

Applying the Cauchy-Schwarz inequality to each inner product, $|\langle a_i, b_j\rangle|^2 \leq \|a_i\|_2^2\|b_j\|_2^2$:

$$\|AB\|_F^2 \leq \sum_{i=1}^{m}\sum_{j=1}^{p}\|a_i\|_2^2\|b_j\|_2^2 = \left(\sum_{i=1}^{m}\|a_i\|_2^2\right)\left(\sum_{j=1}^{p}\|b_j\|_2^2\right).$$

Because the sum of squared row norms of A equals $\|A\|_F^2$, and the sum of squared column norms of B equals $\|B\|_F^2$:

$$\|AB\|_F^2 \leq \|A\|_F^2\|B\|_F^2 \implies \|AB\|_F \leq \|A\|_F\|B\|_F.$$

4. Application to the given matrix:

1. **Direct calculation:**

$$\|A\|_F = \sqrt{1^2+2^2+2^2+4^2} = \sqrt{1+4+4+16} = \sqrt{25} = 5.$$

2. **Singular values and verification:**

Notice that $A = uu^\top$ where $u = \begin{pmatrix}1\\2\end{pmatrix}$. We compute $A^\top A = A^2$:

$$A^\top A = (uu^\top)(uu^\top) = u(u^\top u)u^\top = \|u\|_2^2(uu^\top) = (1^2+2^2)A = 5A.$$

The eigenvalues of $A^\top A$ are roots of its characteristic polynomial:

$$A^\top A = 5\begin{pmatrix}1 & 2\\ 2 & 4\end{pmatrix} = \begin{pmatrix}5 & 10\\ 10 & 20\end{pmatrix}.$$

Since $\text{rank}(A^\top A) = 1$, one eigenvalue is zero: $\lambda_2(A^\top A) = 0$.

The non-zero eigenvalue is given by the trace:

$$\lambda_1(A^\top A) = \text{Tr}(A^\top A) = 5 + 20 = 25.$$

The singular values are the square roots of the eigenvalues of $A^\top A$:

$$\sigma_1(A) = \sqrt{25} = 5, \quad \sigma_2(A) = \sqrt{0} = 0.$$

We verify:

$$\sqrt{\sigma_1(A)^2 + \sigma_2(A)^2} = \sqrt{5^2+0^2} = 5 = \|A\|_F.$$

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Exercise 5.4 (Schatten p-Norms and Singular Value Inequalities)

Let $A \in \mathcal{M}_{m,n}(\mathbb{R})$ with rank $r = \text{rank}(A) \geq 1$. Let $\sigma(A) = (\sigma_1(A), \ldots, \sigma_r(A)) \in \mathbb{R}^r$ denote its non-zero singular values arranged in non-increasing order: $\sigma_1(A) \geq \sigma_2(A) \geq \cdots \geq \sigma_r(A) > 0$.

For any $p \in [1, \infty)$, the Schatten p-norm of A is defined as the ℓ_p-norm of its singular values:

$$\|A\|_{S_p} := \|\sigma(A)\|_{\ell_p} = \left(\sum_{i=1}^{r}\sigma_i(A)^p\right)^{1/p},$$

and for $p = \infty$:

$$\|A\|_{S_\infty} := \|\sigma(A)\|_{\ell_\infty} = \sigma_1(A).$$

1. **Identify the classical matrix norms corresponding to $p = 1$, $p = 2$, and $p = \infty$.**

2. **Using the properties of vector ℓ_p-norms, prove the following chain of inequalities:**

$$\|A\|_{S_\infty} \leq \|A\|_{S_2} \leq \|A\|_{S_1} \leq \sqrt{r}\|A\|_{S_2} \leq r\|A\|_{S_\infty}.$$

   Under what condition on the singular values does equality hold for each bound?

3. **Recall von Neumann's trace inequality: for any $A, B \in \mathcal{M}_{m,n}(\mathbb{R})$,**

$$|\langle A, B\rangle| = |\operatorname{Tr}(A^\top B)| \leq \sum_{i=1}^{\min(m,n)} \sigma_i(A)\sigma_i(B).$$

   Show that for any conjugate exponents $p, q \in [1, \infty]$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$:

$$|\langle A, B\rangle| \leq \|A\|_{S_p}\|B\|_{S_q}.$$

4. **Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.**

   (a) **Compute the eigenvalues of $A^\top A$ and deduce the singular values of A.**

   (b) **Compute explicitly the nuclear norm $\|A\|_{S_1}$, the Frobenius norm $\|A\|_{S_2}$, and the spectral norm $\|A\|_{S_\infty}$.**

   (c) **Verify numerically the chain of inequalities established in question 2.**

****

Solution of the exercise 5.4 1. Classical matrix norms:

- For $p = 1$: $\|A\|_{S_1} = \sum_{i=1}^r \sigma_i(A) = \|A\|_*$ is the **nuclear norm** (also known as the trace norm).
- For $p = 2$: $\|A\|_{S_2} = \sqrt{\sum_{i=1}^r \sigma_i(A)^2} = \|A\|_F$ is the **Frobenius norm**.
- For $p = \infty$: $\|A\|_{S_\infty} = \sigma_1(A) = \|A\|_2$ is the **spectral norm** (operator norm induced by the Euclidean vector norm).

2. Chain of inequalities:

Let $\sigma = (\sigma_1, \ldots, \sigma_r) \in \mathbb{R}^r$ with $\sigma_i > 0$.

- *Inequality* $\|A\|_{S_\infty} \leq \|A\|_{S_2}$*:*
  By definition:

$$\|A\|_{S_\infty}^2 = \sigma_1^2 \leq \sum_{i=1}^r \sigma_i^2 = \|A\|_{S_2}^2.$$

  Taking the square root gives $\|A\|_{S_\infty} \leq \|A\|_{S_2}$. Equality holds if and only if $\sigma_2 = \cdots = \sigma_r = 0$, meaning $\operatorname{rank}(A) = 1$.

- *Inequality* $\|A\|_{S_2} \leq \|A\|_{S_1}$*:*
  Expanding the square of the nuclear norm:

$$\|A\|_{S_1}^2 = \left(\sum_{i=1}^{r} \sigma_i\right)^2 = \sum_{i=1}^{r} \sigma_i^2 + \sum_{i \neq j} \sigma_i \sigma_j = \|A\|_{S_2}^2 + \sum_{i \neq j} \sigma_i \sigma_j.$$

  Since $\sigma_i > 0$, the cross terms are non-negative, so $\|A\|_{S_2}^2 \leq \|A\|_{S_1}^2$, which yields $\|A\|_{S_2} \leq \|A\|_{S_1}$. Equality holds if and only if $r = 1$.

- *Inequality* $\|A\|_{S_1} \leq \sqrt{r}\|A\|_{S_2}$*:*
  Applying the Cauchy-Schwarz inequality to the vectors $\sigma \in \mathbb{R}^r$ and $\mathbf{1} = (1, \ldots, 1) \in \mathbb{R}^r$:

$$\|A\|_{S_1} = \sum_{i=1}^{r} \sigma_i \cdot 1 \leq \sqrt{\sum_{i=1}^{r} 1^2} \sqrt{\sum_{i=1}^{r} \sigma_i^2} = \sqrt{r}\|A\|_{S_2}.$$

  Equality holds if and only if σ is collinear to $\mathbf{1}$, i.e., all singular values are equal: $\sigma_1 = \cdots = \sigma_r$.

- *Inequality* $\sqrt{r}\|A\|_{S_2} \leq r\|A\|_{S_\infty}$*:*
  Since $\sigma_i \leq \sigma_1$ for all $i \in \{1, \ldots, r\}$:

$$\|A\|_{S_2} = \sqrt{\sum_{i=1}^{r} \sigma_i^2} \leq \sqrt{\sum_{i=1}^{r} \sigma_1^2} = \sqrt{r\sigma_1^2} = \sqrt{r}\sigma_1 = \sqrt{r}\|A\|_{S_\infty}.$$

  Multiplying both sides by $\sqrt{r}$ yields $\sqrt{r}\|A\|_{S_2} \leq r\|A\|_{S_\infty}$. Equality holds if and only if $\sigma_1 = \cdots = \sigma_r$.

Combining these four bounds produces the desired chain.

3. Schatten Hölder inequality:

By von Neumann's trace inequality:

$$|\langle A, B \rangle| \leq \sum_{i=1}^{\min(m,n)} \sigma_i(A)\sigma_i(B).$$

Applying the classical Hölder inequality to the non-negative vectors $\sigma(A)$ and $\sigma(B)$ in $\mathbb{R}^{\min(m,n)}$ with $\frac{1}{p} + \frac{1}{q} = 1$:

$$\sum_{i=1}^{\min(m,n)} \sigma_i(A)\sigma_i(B) \leq \left(\sum_{i=1}^{\min(m,n)} \sigma_i(A)^p\right)^{1/p} \left(\sum_{i=1}^{\min(m,n)} \sigma_i(B)^q\right)^{1/q} = \|A\|_{S_p}\|B\|_{S_q}.$$

When $p = 1$ and $q = \infty$, this yields the fundamental matrix duality bound:

$$|\operatorname{Tr}(A^\top B)| \leq \|A\|_* \|B\|_2.$$

4. Application to the given matrix:

1. **Singular values of A:**
   We compute the Gram matrix $A^\top A$:

$$A^\top A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

   The characteristic polynomial of $A^\top A$ is:

$$P(\lambda) = \det(A^\top A - \lambda I_2) = (1-\lambda)(2-\lambda) - 1 = \lambda^2 - 3\lambda + 1.$$

   The discriminant is $\Delta = (-3)^2 - 4(1)(1) = 5$, giving the eigenvalues:

$$\lambda_1 = \frac{3+\sqrt{5}}{2}, \quad \lambda_2 = \frac{3-\sqrt{5}}{2}.$$

   Observe that $\left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}$ and $\left(\frac{\sqrt{5}-1}{2}\right)^2 = \frac{6-2\sqrt{5}}{4} = \frac{3-\sqrt{5}}{2}$. Taking square roots, the singular values are:

$$\sigma_1(A) = \frac{\sqrt{5}+1}{2}, \quad \sigma_2(A) = \frac{\sqrt{5}-1}{2}.$$

2. **Computation of the norms:**

   - **Spectral norm ($p = \infty$):**

$$\|A\|_{S_\infty} = \sigma_1(A) = \frac{1+\sqrt{5}}{2} \approx 1.618.$$

   - **Frobenius norm ($p = 2$):**

$$\|A\|_{S_2} = \sqrt{\lambda_1 + \lambda_2} = \sqrt{\mathrm{Tr}(A^\top A)} = \sqrt{1+2} = \sqrt{3} \approx 1.732.$$

   - **Nuclear norm ($p = 1$):**

$$\|A\|_{S_1} = \sigma_1(A) + \sigma_2(A) = \frac{\sqrt{5}+1}{2} + \frac{\sqrt{5}-1}{2} = \sqrt{5} \approx 2.236.$$

3. **Verification of the inequalities ($r = 2$):**
   Evaluating each quantity:

$$\|A\|_{S_\infty} = \frac{1+\sqrt{5}}{2} \approx 1.618,$$
$$\|A\|_{S_2} = \sqrt{3} \approx 1.732,$$
$$\|A\|_{S_1} = \sqrt{5} \approx 2.236,$$
$$\sqrt{2}\|A\|_{S_2} = \sqrt{6} \approx 2.449,$$
$$2\|A\|_{S_\infty} = 1+\sqrt{5} \approx 3.236.$$

   Since $1.618 < 1.732 < 2.236 < 2.449 < 3.236$, the chain of inequalities holds strictly:

$$\|A\|_{S_\infty} < \|A\|_{S_2} < \|A\|_{S_1} < \sqrt{2}\|A\|_{S_2} < 2\|A\|_{S_\infty}.$$

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Exercise 5.5 (Low-Rank Matrix Completion and Determinantal Variety)
Consider the partially observed 3×3 matrix:

$$M(x,y) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & x & 4 \\ 3 & 6 & y \end{pmatrix},$$

where $x, y \in \mathbb{R}$ are unknown parameters to be determined.

1. **Show that for any choice of $(x,y) \in \mathbb{R}^2$,** $\operatorname{rank}(M(x,y)) \geq 2$**.**
2. **Compute** $\det(M(x,y))$ **in factored form.**
3. **Deduce the minimal rank that $M(x,y)$ can achieve over $\mathbb{R}$, and determine the set of all parameter pairs (x,y) that attain this minimal rank.**
4. **For each family of solutions found in question 3, provide an explicit rank-2 factorization $M(x,y) = UV^\top$ where $U \in \mathbb{R}^{3\times 2}$ and $V \in \mathbb{R}^{3\times 2}$.**
5. **Now suppose that the $(1,3)$ entry is also unknown, i.e., consider:**

$$\widetilde{M}(x,y,z) = \begin{pmatrix} 1 & 2 & z \\ 2 & x & 4 \\ 3 & 6 & y \end{pmatrix}.$$

Determine the minimal rank of $\widetilde{M}(x,y,z)$ over $\mathbb{R}$, and find all triplets (x,y,z) achieving it.

****

Solution of the exercise 5.5 1. Lower bound on the rank:
The rank of a matrix is bounded below by the rank of any of its submatrices. Consider the 2×2 submatrix formed by rows $\{1,2\}$ and columns $\{1,3\}$:

$$S = \begin{pmatrix} M_{11} & M_{13} \\ M_{21} & M_{23} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 4 \end{pmatrix}.$$

Its determinant is:

$$\det(S) = 1 \times 4 - 1 \times 2 = 2 \neq 0.$$

Because this fully observed submatrix is invertible, its two columns are linearly independent. Consequently:

$$\operatorname{rank}(M(x,y)) \geq \operatorname{rank}(S) = 2 \quad \text{for all } (x,y) \in \mathbb{R}^2.$$

2. Factored computation of the determinant:
We perform the elementary column operation $C_2 \leftarrow C_2 - 2C_1$, which preserves the determinant:

$$\det(M(x,y)) = \det \begin{pmatrix} 1 & 2-2(1) & 1 \\ 2 & x-2(2) & 4 \\ 3 & 6-2(3) & y \end{pmatrix} = \det \begin{pmatrix} 1 & 0 & 1 \\ 2 & x-4 & 4 \\ 3 & 0 & y \end{pmatrix}.$$

Expanding along the second column:

$$\det(M(x,y)) = (x-4)(-1)^{2+2}\det\begin{pmatrix}1 & 1\\ 3 & y\end{pmatrix} = (x-4)(y-3).$$

3. Minimal rank and solution set:

Since $M(x,y) \in \mathcal{M}_3(\mathbb{R})$, its rank is strictly less than 3 if and only if $\det(M(x,y)) = 0$:

$$\det(M(x,y)) = 0 \iff (x-4)(y-3) = 0 \iff x = 4 \quad \text{or} \quad y = 3.$$

Combined with the lower bound $\operatorname{rank}(M(x,y)) \geq 2$ from question 1, the minimal achievable rank is:

$$r_{\min} = 2.$$

The set of all parameters attaining this minimal rank is the union of two affine lines in $\mathbb{R}^2$:

$$\mathcal{S} = \left\{(4,y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\right\} \cup \left\{(x,3) \in \mathbb{R}^2 \mid x \in \mathbb{R}\right\}.$$

The corresponding set of completed minimal-rank matrices is:

$$\mathcal{M}_{\min} = \left\{ \begin{pmatrix}1 & 2 & 1\\ 2 & 4 & 4\\ 3 & 6 & y\end{pmatrix} \;\middle|\; y \in \mathbb{R} \right\} \cup \left\{ \begin{pmatrix}1 & 2 & 1\\ 2 & x & 4\\ 3 & 6 & 3\end{pmatrix} \;\middle|\; x \in \mathbb{R} \right\}.$$

4. Explicit rank-2 factorizations:

- **Branch 1 ($x = 4$):** The second column is twice the first column ($C_2 = 2C_1$). Columns 1 and 3 form a basis of the column space:

$$M(4,y) = \begin{pmatrix}1 & 2 & 1\\ 2 & 4 & 4\\ 3 & 6 & y\end{pmatrix} = \begin{pmatrix}1 & 1\\ 2 & 4\\ 3 & y\end{pmatrix}\begin{pmatrix}1 & 2 & 0\\ 0 & 0 & 1\end{pmatrix}.$$

- **Branch 2 ($y = 3$):** The third row is three times the first row ($R_3 = 3R_1$). Rows 1 and 2 form a basis of the row space:

$$M(x,3) = \begin{pmatrix}1 & 2 & 1\\ 2 & x & 4\\ 3 & 6 & 3\end{pmatrix} = \begin{pmatrix}1 & 0\\ 0 & 1\\ 3 & 0\end{pmatrix}\begin{pmatrix}1 & 2 & 1\\ 2 & x & 4\end{pmatrix}.$$

5. Extension to three unknown entries:

Since $\widetilde{M}$ contains non-zero entries (e.g., $\widetilde{M}_{11} = 1$), $\operatorname{rank}(\widetilde{M}) \geq 1$.

A non-zero matrix has rank 1 if and only if all of its rows are scalar multiples of its first row $R_1 = (1,2,z)$:

- For Row 2: $R_2 = \alpha R_1$. Matching the first entry gives $\alpha \times 1 = 2 \implies \alpha = 2$. Therefore:

$$(2,x,4) = 2(1,2,z) = (2,4,2z) \implies x = 4 \quad \text{and} \quad 2z = 4 \implies z = 2.$$

- For Row 3: $R_3 = \beta R_1$. Matching the first entry gives $\beta \times 1 = 3 \implies \beta = 3$. Therefore:

$$(3, 6, y) = 3(1, 2, z) = 3(1, 2, 2) = (3, 6, 6) \implies y = 6.$$

Hence, the minimal rank of $\widetilde{M}(x, y, z)$ is 1, and it is achieved uniquely at $(x, y, z) = (4, 6, 2)$, yielding the outer product:

$$\widetilde{M}(4, 6, 2) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 2 \end{pmatrix}.$$